CAN IT ACTUALLY WORK? SYNTHESIZING FUNCTIONAL CAD MECHANISMS

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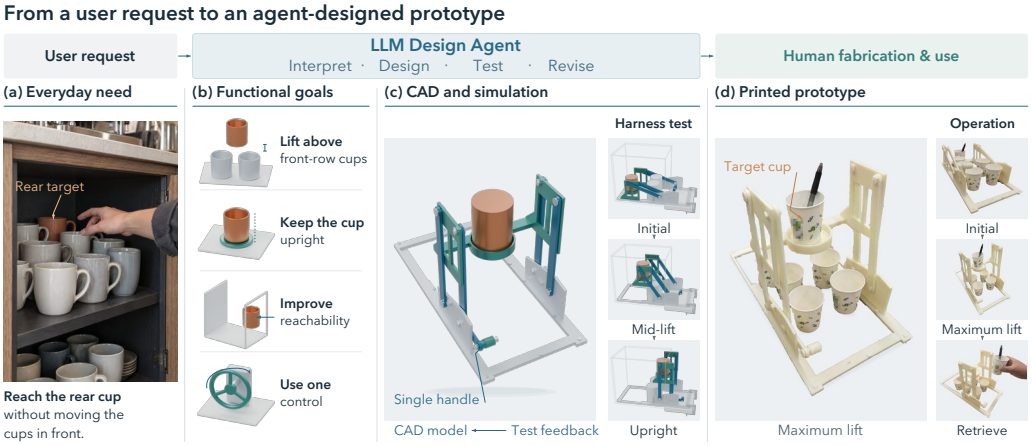


Figure 1: **Functional CAD Design: from a user request to a physical prototype.** A cup-access request is decomposed into functional goals that guide agent-based CAD construction and simulation-driven revision. The resulting design is fabricated and operated by a human; the printed prototype is shown at its maximum lift and during cup retrieval.

ABSTRACT

LLM agents can construct sophisticated CAD models, but visually plausible assemblies need not fulfill their intended functions. We study *functional CAD design*, in which agents synthesize and revise CAD mechanisms to satisfy functional requirements in physics simulation. The same requirement can be realized by different mechanisms, so testing must account for each candidate's controls, components, and operating sequence. To study this problem, we introduce FunCAD-Harness, a harness that connects functional requirements to candidate-specific simulation tests and uses their execution results to guide design revision. We also introduce FunCADBench, spanning 46 mechanism families with dependent and repeated operations, and evaluate frontier agents and baseline workflows. Our experiments show that simulation-guided revision through our harness exposes mechanical failures that are difficult to detect visually and improves functional task success. Together, our benchmark and harness take a step toward advancing LLM agents from geometric modeling to the design of mechanisms that address practical needs.

1 INTRODUCTION

A longstanding ambition for generative AI is to turn human intentions into machines that can be built and made to work. Recent progress in LLM agents brings this ambition closer, enabling them to write modeling programs and operate CAD software to construct sophisticated mechanical designs for fabrication by 3D printing or CNC machining. Yet a visually plausible CAD model does not ensure that the resulting machine will fulfill the user's request: its function depends on how its parts interact

physically during operation. Even an assembly with valid parts and plausible articulation can fail to transmit a load, maintain contact, or return to a state that permits the next operation. A gripper, for example, may grasp an object successfully but leave it poorly positioned after release, preventing the next grasp. Because fabricating and assembling each candidate is costly, physical simulation offers a practical way to examine these interactions under specified inputs and loads, providing feedback for a cycle of design, verification, and redesign.

Verifying a design through physical simulation, however, requires tests that reflect both the user's intended function and the way the candidate mechanism realizes it. Users often describe what they want a machine to accomplish without specifying all the behaviors needed to achieve it. Even after these requirements are recovered, they do not uniquely determine an experiment, because the same function can be realized by mechanisms that operate differently. For an alarm that must trigger autonomously at a requested time, a gravity-powered design requires raising a weight, whereas a mainspring-powered design requires winding. A test designed around one mechanism can therefore be inappropriate for the other, even though the intended function is unchanged.

We study whether LLM agents can interpret underspecified requests and construct and revise CAD mechanisms to satisfy functional requirements in simulation, a problem we term *functional CAD design*. This task requires agents to determine what the requested functions entail, how they can be tested, and how the mechanism should be designed to fulfill them. To examine how far current agents can go on this task, we introduce FunCADBench, a benchmark spanning 46 task families. These tasks range from clamps and transmissions to mechanical counters, alarm clocks, and coin-sorting devices, whose functions depend on different combinations of geometry, force transmission, and changing contacts. Each task specifies a fixed request and an abstract functional test template, while leaving the mechanical realization open. FunCADBench evaluates whether the resulting assembly performs the complete requested behavior and satisfies the applicable geometric and physical constraints. Since repeated physical simulation can be costly for complex mechanisms, we develop a customized simulation environment to support efficient benchmark evaluation.

For this specific task, we develop FunCAD-Harness, which translates user requests into task requirements and abstract physical test templates, and uses structured simulation feedback to guide design revisions. In particular, the harness dynamically determines test conditions appropriate to each candidate mechanism, allowing the intended functions to be verified through physical simulation. Fig. 1 illustrates this workflow through an open-ended cup-access design and its printed prototype.

With FunCAD-Harness, we conduct experiments on FunCADBench using multiple frontier models and baseline workflows. Our experiments show that structured physical testing and diagnostic feedback improve functional task success. In addition, we find that strong visual 3D modeling capabilities do not necessarily translate into superior functional design. Interestingly, when solving these tasks, agents rarely inspect rendered images but instead obtain geometric and physical information primarily through CAD programs and simulation. Together, our benchmark and harness take a step toward advancing LLM agents from geometric modeling to mechanism design that address practical needs.

In summary, our contributions are:

- We formulate *functional CAD design*, in which agents translate underspecified requests into working mechanisms, and introduce FunCADBench, spanning 46 mechanism families, together with a simulation environment for efficient physical evaluation.
- We develop FunCAD-Harness, which derives requirements and abstract test templates from user requests, constructs physical tests suited to candidate mechanisms, and uses structured simulation feedback to guide iterative design and revision.
- We systematically evaluate frontier agents and baseline workflows, demonstrating the benefits of physical testing and structured feedback. Our analyses further reveal performance–cost tradeoffs and recurring challenges in functional mechanical design.

2 Related Work

Mechanical design and AI for engineering. Mechanical design connects intended functions to the structure and motion of mechanisms (Reuleaux, 2013). Computational approaches support this process through topology optimization, performance prediction, and generative design (Bendsøe et al., 2004; Umetani & Bickel, 2018; Regenwetter et al., 2022), while robot co-design jointly considers

morphology and control (Ha et al., 2016; Carlone & Pinciroli, 2019; Yuan et al., 2022). LLMs provide a natural-language interface to design specifications, candidate generation, and engineering analysis (Makatura et al., 2023). Recent systems bring these capabilities into executable design workflows: EngDesign evaluates designs across engineering domains using task-specific simulations (Guo et al., 2025), and VLMgineer combines vision-language models with evolutionary search to co-design robotic tools and their actions (Gao et al., 2026). Closest to our setting, BesiegeField studies machine construction from textual functional demands, using standardized components and simulation-derived rewards and state feedback to support iterative revision (Zhang et al., 2026).

CAD generation and benchmarks. Programmatic representations express 3D geometry through modeling operations and part composition (Ritchie et al., 2023), providing a natural interface for generation with language models. Recent work explores CAD scripts (Wu et al., 2023; Alrashedy et al., 2025; Li et al., 2025), Blender macros (Huang et al., 2024), and symbolic graphics programs (Qiu et al., 2025; Chen et al., 2025). CADPrompt pairs natural-language descriptions with reference CAD code, while CADCodeVerify uses rendered views to identify and correct deviations in generated geometry (Alrashedy et al., 2025). In our setting, agents design both part geometry and assembly structure to realize functional requests in CAD models intended for fabrication. FunCADBench evaluates these mechanisms through simulation of assemblies constructed from their source CAD geometry, with success requiring the complete requested behavior, including dependent and repeated operations.

Agents and harness design. LLM agents use tools and iterative feedback to revise their outputs (Schick et al., 2023; Shinn et al., 2023; Madaan et al., 2023), while multi-agent frameworks distribute generation and critique across roles (Li et al., 2023; Chen et al., 2024). Beyond revising artifacts, language models can also construct executable objectives: Eureka generates reward functions from task descriptions and environment code, refining them through feedback from policy training (Ma et al., 2023). In FunCAD-Harness, the mechanism itself changes during design, so an independent testing agent adapts physical tests to each candidate while retaining the requested functions. Executing these tests supplies feedback for subsequent structural revision.

3 FunCADBench: Benchmark for Functional CAD Design

3.1 Functional CAD design

Mechanical design involves production cost, manufacturing processes, and regulatory requirements as well as mechanical behavior (Pahl et al., 2007; Nisbett & Budynas, 2020). We focus on interpreting high-level user requests and determining whether a CAD mechanism fulfills the intended functions under appropriate physical conditions (Zhang et al., 2026). Formally speaking, each task of functional CAD design is specified by a natural-language *request* describing a mechanism's intended use. The *request* establishes its *functional requirements*, which describe the behaviors to achieve, and *fixed constraints*, which must be preserved throughout design. The agent produces a *candidate design* together with *declared physical assumptions*. Different mechanical implementations are acceptable as long as they satisfy these requirements; for example, a clamp may use different actuation mechanisms, but it must still close, hold an object, and release it.

3.2 Task Families and Taxonomy

To evaluate frontier LLM agents on this task across different mechanical demands, we introduce FunCADBench, a benchmark spanning 46 mechanism families (representative mechanisms are shown in Fig. 2). These range from elementary assemblies to coupled transmissions and devices with sequential or repeated operation. Fig. 3 shows mechanisms that LLM agents built for all 46 families in our experiments. Many of these families are drawn from familiar artifacts, such as clamps, whose intended uses can be described without specifying their internal mechanics. Several families pose open-ended mechanism design tasks that leave key architectural choices to the constructor. For example, the four-station indexing table (Task 40) must convert continuous input rotation into intermittent motion with dwell, without prescribing the indexing mechanism. We choose these families to cover different combinations of challenges along these axes–motion primitives, topology, complexity, contact, and geometric difficulty–each drawing on different agent capabilities (Table 1).

Table 1: Task taxonomy of FunCADBench.

Axis	Mechanical demand	Relevant agent capabilities
Motion primitives	Motion conversion, transmission, phase, and repeated operation	Spatial and physical reasoning about motion
Topology	Connections, branches, and kinematic loops	Reasoning about assembly constraints
Complexity	Coupled subsystems, states, and operation sequences	Task decomposition and planning across revisions
Contact	Engagement, sliding, locking, and friction	Reasoning about local physical interactions
Geometric difficulty	Feature construction, alignment, clearance, and assembly access	CAD tool use and geometric verification

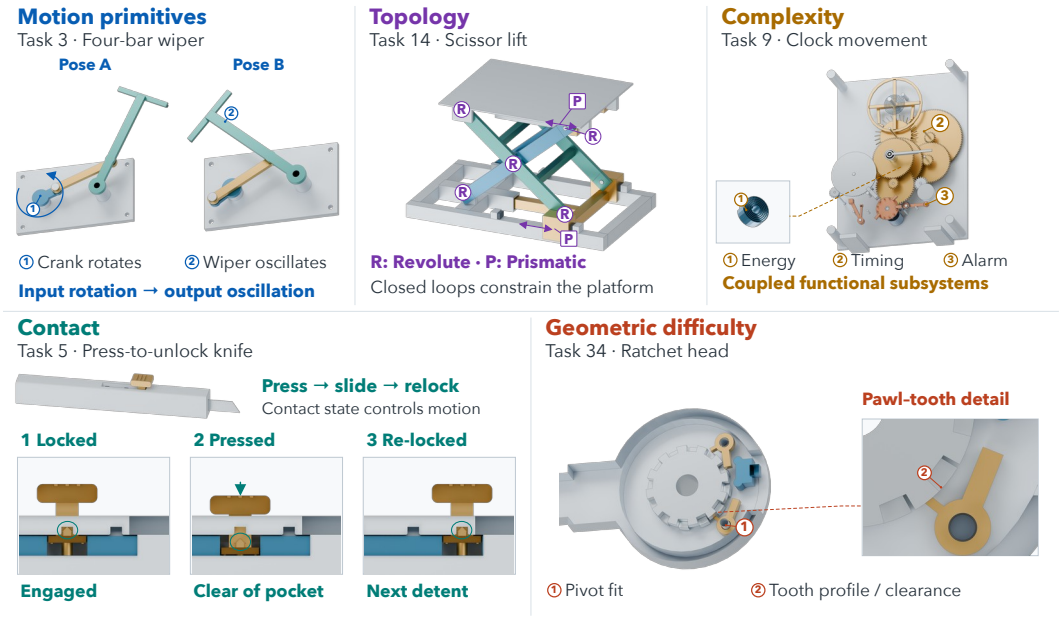


Figure 2: Taxonomy with representative mechanisms. Colors indicate functional parts and subsystems.

3.3 Physical Evaluation and Success Criteria

Functional requirements specify what an assembly must accomplish, while its mechanical realization determines how those functions can be tested. Following the separation between test logic and implementation-specific parameters in ProjectEval (Liu et al., 2025), we define abstract test templates for the required behaviors and bind them to each candidate mechanism. For registered benchmark tasks, the requests and templates remain fixed across candidates. Binding identifies the candidate's operating interfaces and observation signals and specifies the inputs, loads, physical attributes, and environmental conditions needed to execute the template. These choices follow the mechanism while retaining the task's required functions and fixed constraints. Operating or performance parameters left open by the task may be selected or revised within its permitted scope; such adjustments are recorded separately from the mapping of operations and measurements (Appendix A.4).

Physical execution records forces, positions, and contact or collision events for the relevant mechanical modules. Explicit checks of these signals against the requirements determine which functional checks pass and establish which functions have been demonstrated. Each result retains this evidence together with the operating conditions and physical modeling assumptions used in the tests.

We measure end-to-end functional-design completion: whether a workflow constructs an assembly and completes its verification under the task's simulation protocol. A task is completed when the final delivered assembly passes every required functional check and all applicable source-geometry checks under its declared operating conditions. The same completion rule applies to all workflows, including baselines whose assemblies are adapted into executable physical models after construction. Incomplete preparation or execution counts as non-completion, with its status distinguished from an observed mechanical failure. Dependent operations are tested from the states produced by preceding



Figure 3: **Mechanisms built by LLM agents in our experiments.** The 75 assemblies cover all 46 FunCAD-Bench families and are rendered from their delivered CAD geometry (arranged for display, not to scale); 38 of them pass the full FunCAD-Harness evaluation.

operations: a balance's recovery, for example, starts from its state under unequal loading. Independent functions may be tested separately, with all tests evaluating the same delivered configuration. Per-function results distinguish passed checks, observed mechanical failures, and unexecuted or inconclusive tests.

3.4 Simulation Environment

Repeated design revision requires physical tests that can be executed within an agent's working budget. Our execution environment uses Project Chrono for rigid-body dynamics and Bullet for collision detection. It also supports equivalent models of selected mechanical interactions. Gear couplings, for example, represent transmission between rotating components without resolving individual tooth contacts (Coros et al., 2013; Tasora et al., 2016).

The constructor and test-binding agent select native simulation or a declared equivalent model for each interaction and record its parameters and assumptions. Equivalent models represent mechanical interactions through force or torque laws applied to the corresponding source components, while the native solver computes the resulting rigid-body motion. These models specify interaction laws rather than prescribing output trajectories. Connections are associated with finite regions of the source geometry and are released when those regions separate. Functional checks evaluate the simulated motion, while separate audits inspect the original CAD geometry at sampled poses. Appendix D details the interaction models, audit scope, and execution procedure.

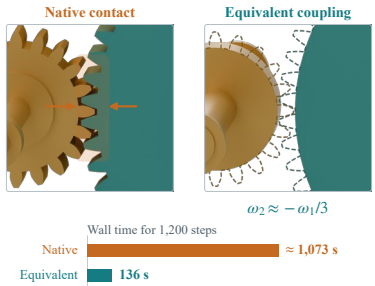


Figure 4: **Gear-transmission cost.** Native time is extrapolated from its recorded mean.

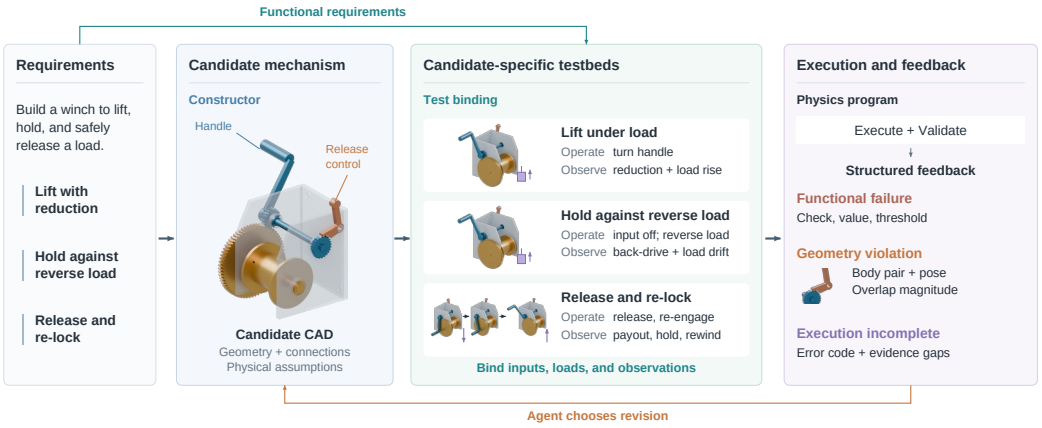


Figure 5: Overview of FunCAD-Harness, illustrated with a releasable reduction winch (Task 42).

4 Agentic Harness for Functional CAD Design

FunCAD-Harness connects requirement specification, CAD construction, physical testing, and revision in a closed loop (Fig. 5). Building on agentic CAD generation (Zhou et al., 2026) and simulation-based machine design (Zhang et al., 2026), the harness coordinates a constructor that develops the mechanism and a separate test-binding agent that binds functional test templates to each candidate's operating inputs and observable states that constructs physical tests for each candidate. These tests connect the requested functions to the candidate's actual mechanics, and their execution provides feedback for the next design revision.

4.1 Functional requirements and test templates

Registered FunCADBench tasks use fixed requests and functional test templates across candidate designs. Each template specifies the operations, observations, and success criteria for the required behavior, leaving mechanism-dependent details to candidate-specific binding. For requests outside the registered benchmark, the constructor first identifies required functions, fixed constraints, and open operating parameters, then proposes a test template. This template-authoring step is separate from the candidate-specific binding used to evaluate the 46 registered benchmark tasks. A separate test-binding agent checks its coverage and executability, and the constructor revises it accordingly. Requirements describe intended behavior: for an alarm clock, the alarm must trigger autonomously at the requested time, rather than require a particular escapement design.

4.2 CAD construction and physical testing

The constructor develops the CAD geometry, connections, operating inputs, and physical assumptions. Candidate-specific test binding connects each template to the delivered mechanism while preserving its geometry and declared mechanics. For the winch example (Fig. 5), binding identifies the handle, release control, and load attachment and configures tests of lifting, holding, release, and re-locking. Test parameters come from the task, geometry, design documentation, or task-authorized constructor choices. The binding specifies the physical experiment; the execution program determines its outcome from the recorded signals and applicable geometry checks. Parameter provenance and permitted adjustments are retained as described in Appendix A.4.

4.3 Revision from physical feedback

The harness executes the bound tests without changing the CAD design between tests and checks its original geometry at sampled motion poses (Section 3.3). It returns per-function results, physical signals, failed checks, geometry violations, and execution errors. The constructor uses this feedback to diagnose failures and revise geometry, connections, permitted operating parameters, or simulation representations. Since a revision may affect several functions, the test-binding agent updates candidate-specific bindings for all required functions, which the harness reruns on the revised assembly.

5 EXPERIMENTS

5.1 FUNCTIONAL DESIGN PERFORMANCE

We evaluate four agent–model configurations—Astra, Kimi, Fable, and Grok—with FunCAD-Harness on all 46 tasks, allowing one independent attempt per family and a three-hour active-time budget. During generation, each agent uses its own model to drive the ReAct loop for CAD construction, testing, and revision. Internal tests and revisions are part of this generation process.

After completing a batch of generation runs, we evaluate the resulting machines in a separate batch using GPT-6 Astra with a one-million-token context allowance and medium reasoning. The same final-evaluation model, context allowance, and reasoning setting are used across all generation models and comparison settings, including Grok + history. The final evaluation agent configures candidate-specific physical tests under the task's required functions and fixed constraints; the execution program determines their outcomes. Appendix E provides the model settings and time-accounting details.

Across the four models, agents solve 34 of the 46 tasks (Fig. 6). Their performance diverges mainly on more demanding mechanisms: all four succeed on most low-complexity tasks, while every task solved exclusively by Fable has medium or high complexity. High-complexity tasks remain challenging even for Fable, the strongest single-attempt model, which solves five of the 15 tasks in this group. Contact coordination has the lowest demand-weighted success rate for every model. We also evaluate Grok with up to eight attempts per task, access to records from earlier attempts. Grok solves 36 of the 46 tasks, including six unsolved by any single-attempt configuration, with only half of the API cost of Astra runs.

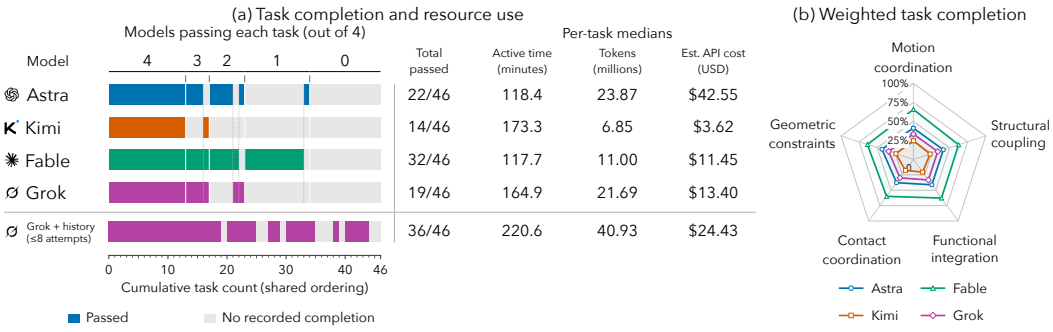


Figure 6: Task success and mechanical demand coverage. (a) Tasks share an ordering across models; colored segments indicate success and gray segments indicate no recorded success. Individual outcomes appear in Table 8. Time, tokens, and estimated constructor API cost are per-task medians. API cost covers recorded constructor usage and accumulates across attempts in the history setting (Appendix E.3). (b) Success weighted by requirement grades 1–3 for each demand axis, normalized by the total weight over all 46 tasks (Appendix E.5). Task grouping and demand-weighted scores use the four single-attempt settings.

5.2 HARNESS COMPARISONS

We compare FunCAD-Harness with three workflows that vary the support for physical testing (Table 2). In the *requirements-only* and *simulation-tools* baselines, agents choose their own tests and may implement their own simulations. The *binary-feedback* variant retains the full harness workflow but returns only pass/fail simulation outcomes. Independent evaluation agents assess the final assemblies produced by each workflow.

Simulation access and whole-task success. Providing simulation tools increases task success over requirements alone, and the full harness yields further gains: from 8 to 22 tasks for Astra and from 10 to 19 for Grok. The gains span motion transmission and contact-dependent locking. Both models additionally solve the rack-and-pinion gate, four-bar wiper, and hand-cranked whisk (Tasks 2–4), as well as the self-locking knife and spring-latch lock (Tasks 5 and 33).

Astra tests a clock's escapement for ten simulated minutes and its striker separately (*simulation-tools setting*), but leaves the sequence of setting, triggering, and stopping the alarm while preserving timekeeping unverified. The delivered hour wheel also interferes with the alarm lever (Figure 7b).

Table 2: End-to-end completion on 46 functional CAD tasks. All workflows undergo independent final evaluation. Success-set overlaps appear in Appendix F.

Setting	Provided support	Astra	Grok
Requirements only	Requirements; agent-chosen verification	1/46	3/46
Simulation tools	Requirements, simulator, and tool guidance	8/46	10/46
Full harness	Requirement-level tests and diagnostic feedback	**22/46**	**19/46**
Binary feedback	Full workflow with pass/fail simulation feedback	0/46	8/46

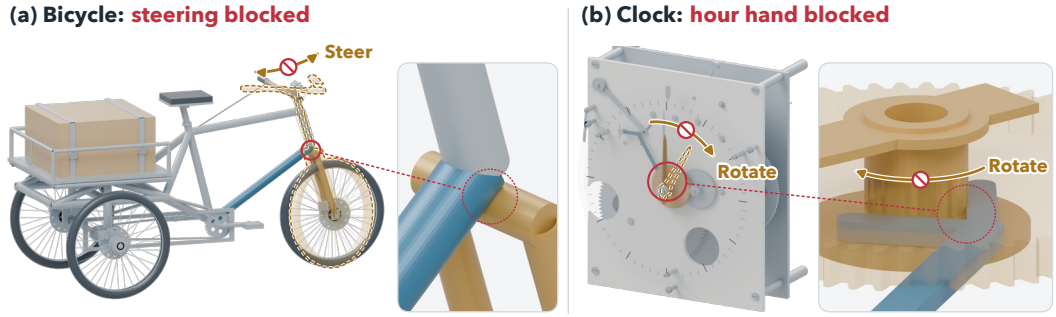


Figure 7: Function-blocking interference in Astra designs produced without harness feedback. (a) The bicycle's steering assembly intersects the fixed frame (Task 7, *Requirements-only*). (b) The clock's hour wheel overlaps the alarm lever despite separate escapement and striker tests (Task 9, *Simulation-tools*). Red circles mark interference; dashed silhouettes indicate intended poses. The clock inset uses schematic hand placement and sleeve length.

Retaining functional checks through revision. Functional checks can also be lost while repairing an execution error. In the simulation-tools setting, Grok replaces a numerically invalid loaded test of a ratchet screwdriver (Task 13) with a short stepping test, leaving the drive and return-stroke requirements unchecked. A separate full-harness run retains the functional checks: four stroke checks pass, while both output-backslip checks fail.

Diagnostic feedback and workflow completion. Replacing diagnostic feedback with pass/fail outcomes reduces recorded success from 22 to zero tasks for Astra and from 19 to eight for Grok. The full harness supplies failed requirements, measured discrepancies, and physical-validity violations that can distinguish failure causes and guide repairs. Astra's binary-feedback runs include early delivery of partial results, so its zero successes include unfinished evaluation and failed checks. Diagnostic feedback improves whole-task completion, supporting both design revision and continued testing.

5.3 Design and Revision Behavior

Repairing failures in repeated operation. Fable's gripper (Task 41) passes a short development test, but formal repeated-use testing reveals that the workpiece tips after release and the next grasp misses it. The second grasp records 0 N against a 10 N minimum. Fable widens the contact pads and support rest and reduces the permitted input-force limit from 50 to 35 N. The revised design leaves the workpiece positioned for reuse: its second grasp reaches 14.40 N under the unchanged 5 N external load, and both grasp–release cycles and the geometry audit pass (Fig. 9).

Functional recovery can still leave geometric interference unresolved. In Grok's spring-return door latch (Task 33), both re-locking checks initially fail because the latch rests on the strike-plate edge after closure, leaving 13 of 15 functional checks passed. Grok reshapes the entry bevel and adds a catching shoulder, restoring engagement after both closures and passing all 15 functional checks. Remaining latch–faceplate interference requires a further local relief of the faceplate. The final revision passes both functional checks and the source-geometry audit under unchanged test parameters (Figure 8).

Verification strategies and execution time. We analyze tool use on 41 task families with observed standard-tool execution time for all four models (Appendix E.6, Fig. 11). Fable and Grok devote larger shares of their calls to information access, whereas Astra devotes more to simulation, geometric checks, and feedback analysis. Astra's share of simulation calls exceeds Fable's on 40 of the 41

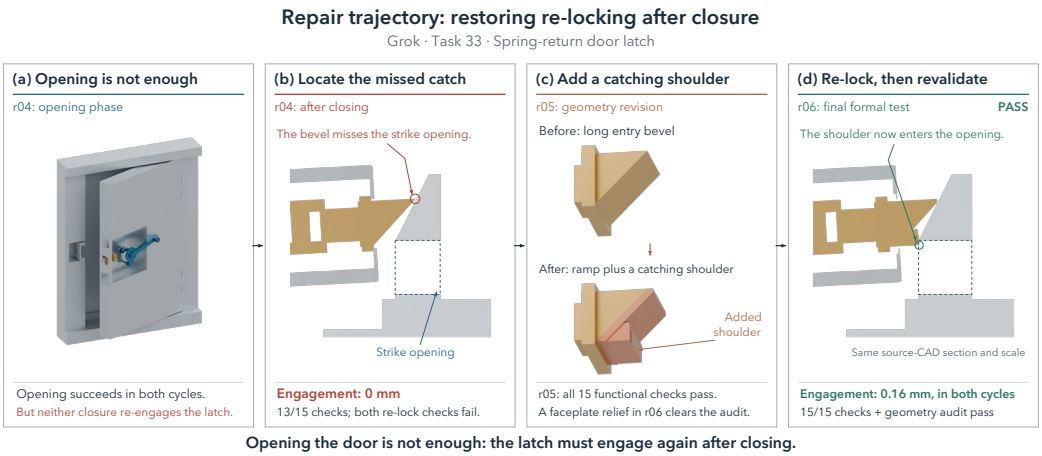


Figure 8: Grok repairs re-locking and local interference in a spring-return door latch (Task 33).

matched tasks. Despite these different action profiles, simulation and cost probes dominate observed standard-tool execution time for every model, accounting for 75–85% on average across tasks.

6 Discussions

From simulation to fabrication. Turning a simulated mechanism into a working machine requires accounting for uncertainty in both physics models and fabrication. Contact and friction approximations, manufacturing tolerances, assembly misalignment, and material variation can change whether parts slide, engage, or jam. Selected prototype experiments could calibrate uncertain parameters and identify missing effects, following simulation calibration from real-world robot trajectories (Chebotar et al., 2019). Testing across these parameter ranges would assess robustness to fabrication and operating conditions. World models offer fast predictions of rigid, articulated, and deformable interactions (Huang et al., 2026; Yang et al., 2025); adapting them with prototype measurements could capture hard-to-model effects and make repeated design tests more affordable.

Coverage of functional tests. Like project-level software generation in ProjectEval (Liu et al., 2025), functional CAD design turns a high-level request into an artifact whose components must work together. In both settings, ambiguous requirements and incomplete test suites can leave intended functions unchecked, while component tests can miss interactions. Mechanical tests must also vary loads, initial states, and operation sequences, which determine contacts and the states available to later operations. Future harnesses could combine requirement review and reusable engineering tests with automated searches for failure cases within the intended operating range (Formica et al., 2024). These tests could guide revisions and preserve verified behavior.

Multimodal interface fluency and physical reasoning. Despite impressive visual 3D reconstruction (Sun et al., 2026), Astra does not dominate functional CAD design. Fable achieves higher single-attempt success, while Grok with additional attempts and retained history surpasses every single-attempt configuration at lower median cumulative constructor API cost than single-attempt Astra. This contrast raises the possibility that visual modeling performance partly reflects multimodal interface fluency, rather than a corresponding advantage in physical reasoning. Targeted post-training can develop such proficiency in computer-use and CAD workflows (Ramrakhya et al., 2025; Gong et al., 2026). Our agents rarely inspect rendered images; they obtain geometric and physical information primarily through CAD programs and simulation. Post-training on these design–test–repair trajectories, together with harnesses that help agents reuse prior designs and feedback, could improve the cost–performance tradeoff in functional design.

7 Conclusion

We introduced *functional CAD design*, which asks agents to turn high-level user requests into CAD mechanisms that perform the intended functions in simulation. To study this problem, we built FunCADBench, a benchmark that includes 46 mechanism families which covers a wide range of

possible mechanisms. We also proposed FunCAD-Harness, a harness for LLM agents that bridge demand prompts to physical tests tailored to each CAD assembly candidate and uses simulation feedback to guide design revision. Our experiments show that our harness substantially improves whole-task success over access to simulation tools alone, enabling agents to repair failures in coupled and repeated operations. We hope that our exploration sheds light on using LLM agents to bridge the gap between 3D generation and real-world mechanical design.

AI USE STATEMENT

AI language-model assistants were used in the preparation of this manuscript to restructure drafts, draft and revise text, and prepare the LaTeX source. The agents studied in this work are themselves large language models; experimental outputs, measurements, and verdicts were produced by the Harness execution pipeline and are reported as-is from recorded traces. All technical claims, numerical results, citations, and figures were verified by the authors against the underlying execution evidence before inclusion. AI assistance was not used to generate, alter, or select experimental results. The authors take full responsibility for the content of the paper.

ETHICS STATEMENT

This study evaluates computational agents on mechanical-design tasks and does not involve human-subject experiments. Automated mechanical design could make engineering tools more accessible, but it may also enable unsafe designs or misuse. Our evaluation establishes functional behavior only under the specified simulation conditions and modeling assumptions; passing these tests does not establish durability, manufacturability, or real-world safety. Generated mechanisms require engineering review and physical validation before practical deployment, particularly in applications where failure could cause injury. The fabricated cup-access prototype demonstrates one design's physical realization and should not be interpreted as validation of all benchmark mechanisms.

REPRODUCIBILITY STATEMENT

The paper documents the experimental settings and evaluation procedures needed to interpret and reproduce our comparisons. Appendix A describes model-visible instructions, tool interfaces, and feedback conditions. Appendices B and C describe the 46 mechanism families, functional requirements, and examples of executable tests. Appendix D details the simulation interface, physical modeling assumptions, and geometry audits. Appendix E reports model and agent configurations, execution budgets, time accounting, and outcome aggregation. Appendix F provides the ablation protocols, task-level results, and supporting behavioral analyses. Reported outcomes are linked to evaluated design revisions, test protocols, and recorded execution evidence, with incomplete evaluations distinguished from established functional success. Exact reruns may vary because of stochastic agent behavior and changes to externally hosted models or agent interfaces; reproductions should therefore report their configurations and preserve the evaluated designs, operating conditions, and execution records.

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Appendix

Table of Contents

A Prompts and Model-Visible Materials **14**
- A.1 Task prompts and constructor instructions . . . 14
- A.2 Harness Tool Descriptions and Returned Information . . . 15
- A.3 Test-Binding Instructions . . . 16
- A.4 Permitted Operating and Performance Adjustments . . . 17
- A.5 Template Authoring and Review for Open-Ended Requests . . . 18
- A.6 Materials for the Comparison Settings . . . 19

B Mechanism Families and Task Taxonomy **19**
- B.1 Mechanism Catalog . . . 19
- B.2 Taxonomy Dimensions and Representative Mechanisms . . . 23

C Functional Specifications and Diagnostic Cases **24**
- C.1 From Input Requirements to Executable Tests . . . 24
- C.2 Worked Example: Designing and Refining a Parallel Gripper . . . 26
- C.3 Fixed Requirements, Adjustable Parameters, and Binding . . . 27
- C.4 State Dependencies and Completion . . . 27
- C.5 Diagnostic Interpretation . . . 27
- C.6 Repair Trajectories . . . 28

D Scene Interface and Physical Execution Details **29**
- D.1 Scene Interface and Initialization . . . 29
- D.2 Contact, Connections, and Equivalent Models . . . 29
- D.3 Trajectory Recording and Geometry Auditing . . . 29

E Agent Configurations, Budgets, and Evaluation Protocol **30**
- E.1 Model and Agent Configurations . . . 30
- E.2 Attempts and Time Accounting . . . 30
- E.3 Artifact Selection and Outcome Aggregation . . . 30
- E.4 Task-Level Completion Results . . . 32
- E.5 Demand-Weighted Completion and Complexity Strata . . . 33
- E.6 Tool-Use Annotation and Execution-Time Profiles . . . 34

F Detailed Harness Comparisons **35**
- F.1 Experimental Settings and Evaluation . . . 36
- F.2 Task-Level Outcomes . . . 37
- F.3 Verification Scope and Revision Without the Harness . . . 38
- F.4 Diagnosis and Stopping Under Binary Feedback . . . 39
- F.5 Construction Without Visual Access . . . 40

G 3D-Printed Mechanical Prototypes **41**

A Prompts and Model-Visible Materials

A.1 Task prompts and constructor instructions

Each of the 46 registered task prompts combines family-specific functional requirements with shared CAD-delivery instructions. The requirements specify the intended use, required functions,

permitted operations, and fixed constraints, while leaving the choice of geometry and assembly structure to the constructor. Attempts on the same registered family use the same task text; additional harness instructions describe the available tools and provide run-specific information, including the execution budget. Experiment-specific task adjustment (operating/performance parameters) are recorded separately from the original registered text. Fig. 5 presents an abbreviated description of the Task 42 winch task, whose functional requirements include lifting with reduction, holding against reverse load, and release followed by re-locking.

The box below summarizes the principal instructions provided to the constructor during the experiments, retaining their design, delivery, and verification requirements while omitting file organization and execution details.

Constructor instructions (summary)

Construct a complete and recognizable CAD assembly that satisfies the required functions and preserves the task's fixed constraints. Include all moving parts, supports, connections, and internal physical effects required for its intended functions.
Provide rebuildable modeling source, a complete STEP assembly, the information needed for physical simulation, and concise operating instructions.
Identify the actual controls and their operating directions, the normal operating sequence, observable physical states, and the units and basis of relevant parameters.
Use the modeling representations permitted by the workflow, and document the sources of their parameters and the limits of their applicability.
Support functional claims with execution evidence; do not substitute prescribed output motion, fabricated measurements, or undeclared fixation for the mechanism's functionality.
Leave formal test binding to the separate test-binding agent, who maps the test requirements to the actual design's components, operations, measurements, and parameters.
Use execution feedback to revise the design, and submit the updated candidate for renewed binding and evaluation.
Do not assign a formal pass verdict yourself; the harness determines completion from execution evidence.

These instructions separate mechanical construction from formal test binding: the constructor delivers the mechanism and its operating instructions, the test-binding agent binds the formal tests, and the program determines their outcomes from execution evidence.

A.2 Harness Tool Descriptions and Returned Information

The harness provides tools for building, previewing, checking, and simulating candidate mechanisms. Formal evaluation combines source preparation, independent test binding, physical execution, and source-geometry auditing.

Source preparation loads the delivered geometry and physical scene for evaluation. After independent binding, interface checks verify that the required inputs and observations are connected to the prepared model; passing these checks does not establish functional success. Physical execution produces the signals used to evaluate functional criteria, complemented by audits of the original geometry at the initial configuration and sampled motion poses.

The following excerpt shows the structured result of the second formal evaluation of the Task 3 wiper. The excerpt retains all five functional checks and the geometry-audit outcome, while omitting assertion expressions and ancillary metadata.

Example structured evaluation result: Task 3


```
{
  "status": "PASS",
  "revision": 2,
  "tests": [{
    "test_id": "continuous_wiping",
    "execution_status": "completed",
    "physical_pass": true,
    "assertions": [
      {"id": "complete_sweeps", "passed": true,
       "observed_operands": [4.0, 3.0]},
      {"id": "links_attached", "passed": true,
       "observed_operands": [1.4331800428598694e-06, 0.0002]},
      {"id": "actual_input_cycles", "passed": true,
       "observed_operands": [4.0, 3.0]},
      {"id": "same_phase_return", "passed": true,
       "observed_operands":
         [3.468221060552712e-05, 0.017453292519943295]},
      {"id": "total_input_progress", "passed": true,
       "observed_operands":
         [28.192713231822275, 18.84955592153876]}
    ],
    "geometry_audit": {"passed": true}
  }],
  "complete_instance_pass": true
}
```


In each `observed_operands` pair, the first value is the measured statistic and the second is its comparison bound. Cycle counts and accumulated rotation must meet or exceed their lower bounds, while joint separation and return error must not exceed their upper bounds. Lengths and angles are reported in metres and radians, respectively. All five functional checks and the source-geometry audit passed, and the harness marked the candidate as complete under the tested conditions.

A.3 Test-Binding Instructions

The final evaluation agent instantiates the task template for the submitted mechanism by identifying its operating inputs, loads, and observable physical states. The constructor's operating instructions inform this process, but the correspondence is checked against the candidate's actual geometry and mechanics. For example, testing a toolbox requires identifying the moving lid, its hinge, and a measurement of its opening angle; component names alone do not establish these relationships.

Test-binding instructions (summary)

Inspect the delivered candidate and map each required function to the components, operating inputs, and measurements needed to test it.
Preserve the delivered geometry and declared mechanics when constructing the test binding.
Specify the driven connections, loading conditions, effort schedules, and observation signals according to the candidate's normal operation.
Record each test parameter together with its value and source, including task requirements, geometric measurements, source documentation, or a declared constructor decision.
Apply permitted parameter adjustments while retaining the original values, revised values, and reasons for the changes.
After each design revision, ensure that the binding and evaluation evidence correspond to the updated candidate. Do not combine checks from different candidate configurations to claim completion of a single design.

The test-binding agent configures the candidate-specific experiment, while the program determines its outcome from execution evidence.

Example of candidate-specific binding. The following excerpt shows the final evaluation agent's binding for the second revision of the Task 3 wiper. Comments marked FILLED identify entries supplied when constructing the executable binding, including numerical parameters and component references. The provenance of these entries is recorded separately. Loading code and additional observation routines are omitted.

Candidate-specific binding example: Task 3

```
PROTOCOL = {
    "period_s": 1.8333333333333333,          # FILLED
    "repeats": 3,                            # FILLED
    "input_speed_rad_s": 6.283185307179586, # FILLED
    "input_torque_nm": 2.0,                  # FILLED
    "gain": 0.02,                            # FILLED
    "sweep_rad": 1.53588974175501,           # FILLED
    "return_tolerance_rad":
        0.017453292519943295,                # FILLED
    "min_input_turn_rad":
        6.283185307179586,                   # FILLED
    "max_joint_separation_m": 0.0002,        # FILLED
}

frame = bodies["frame"]  # FILLED: reference body

model.actuators["input"] = Actuator(
    bodies["crank"],     # FILLED: driven body
    "body_torque",
    reference=frame,
    axis=(0.0, 0.0, 1.0),  # FILLED: drive axis
    react_on_reference=False,
)

model.signals["input.angle_rad"] = Signal(
    bodies["crank"],     # FILLED: observed input body
    "body_angle",
    reference=frame,
)

model.signals["wiper.angle_rad"] = Signal(
    bodies["rocker"],    # FILLED: observed output body
    "body_angle",
    reference=frame,
)
```

The operation and observation names, such as `input` and `wiper.angle_rad`, belong to the fixed template interface. The test-binding agent connects these interfaces to the candidate's actual crank, rocker, and reference frame. The numerical protocol implements the constructor's recorded parameter request, with the provenance of each value retained alongside the binding. Here, `period_s` denotes the observation allowance per required cycle, not the commanded input-shaft period. With three required cycles, the total observation duration is 5.5 seconds.

A.4 Permitted Operating and Performance Adjustments

The task distinguishes required functions and explicit constraints from operating or performance parameters that the constructor may adjust. For example, a clamp task may permit changes to applied load or operating duration while still requiring closing, holding, and release. The constructor selects these values within the task's permitted scope. The test-binding agent incorporates them into the test protocol and records their original values, revised values, and the reason for each change; programmatic validity checks remain in force. Completion is assessed under the final tested conditions, which are reported with the adjustment history.

Example of a constructor-requested adjustment. After the first formal Task 3 evaluation failed its motion checks, the constructor requested an adjustment to the operating protocol. The following example combines the recorded parameter request with its accompanying decision rationale. Only the changed parameters are shown, and the rationale is condensed for presentation.

Constructor-requested adjustment: Task 3

Recorded action: `adjust_task`.

Requested parameter values:

```
{
  "parameters": {
    "period_s": 1.8333333333333333,
    "gain": 0.02
  }
}
```

Preserved targets. Retain the commanded input speed of 60 rpm, the finite torque cap of $2\,\mathrm{N\,m}$, at least three complete cycles, an $88°$ sweep, a maximum same-phase return error of $1°$, and a maximum connection separation of $0.2\,\mathrm{mm}$.

Changes from the previous protocol. Reduce the speed-control gain from 0.8 to $0.02\,\mathrm{N\,m\,s/rad}$. Increase the observation allowance per required cycle from 1.5 to approximately 1.8333 seconds, extending the total observation duration from 4.5 to 5.5 seconds.

Worker rationale, condensed. The constructor attributed the repeated input reversals in the preceding formal trace to excessive speed-control gain. It requested gentler speed correction and a longer observation window.

Recorded trade-off. Accept slower speed correction, greater load-dependent speed error, and a longer observation window while preserving the functional targets and source geometry.

The final evaluation agent implemented both requested changes in the updated binding. The remaining protocol parameters and source CAD geometry were unchanged. The second formal evaluation passed all five functional checks and the source-geometry audit, as illustrated in Appendix A.2. This example documents an operating-parameter adjustment and its subsequent evaluation; it does not isolate the effect of either parameter change individually.

A.5 Template Authoring and Review for Open-Ended Requests

Registered tasks reuse fixed functional templates, whereas open-ended requests require templates to be constructed for the requested functions. The constructor proposes the template, and an final evaluation agent checks it before candidate-specific binding.

Template-authoring instructions (summary)

Decompose the request into distinct functional requirements, each with an observable outcome and an explicit pass criterion.
For each requirement, specify the operating conditions, input sequence, measurements, and parameters to be determined for the candidate mechanism.
Preserve dependencies between operations: when a later function relies on an earlier operation, evaluate it from the physical state produced by that operation.
Define requirements in terms of the behavior the mechanism must achieve, rather than the internal steps of a particular implementation. For a mechanical alarm clock, the alarm triggering at the set time is a functional requirement, whereas the escapement engaging a particular tooth describes an implementation detail.
Reuse the shared execution components and submit the proposed template for independent review.

Template-review instructions (summary)

Check that the template covers the requested functions and preserves the task's fixed constraints.
Verify that the proposed operations, observations, and pass criteria express the intended behavior and can be implemented through the available interfaces.
Return missing or incorrect test definitions to the constructor for revision.
If a required function cannot be evaluated using the available operations or observations, record the limitation and leave that function unverified.

The constructor may revise the template, but template review and formal instance binding remain independent responsibilities. A candidate is not considered complete while any required function remains unverified.

A.6 Materials for the Comparison Settings

The four settings in Table 3 vary the support provided for simulation, functional evaluation, and diagnosis. Requirements only does not prohibit self-written simulations. Simulation tools supplies execution support but does not independently organize or enforce coverage of every required function. Full harness and binary feedback share the requirement-level workflow, differing in the diagnostic simulation feedback returned to the constructor.

Table 3: Model-visible support in the four harness settings.

Setting	Model-visible materials and feedback
Requirements only	Functional requirements and ordinary construction tools; agents may implement their own verification.
Simulation tools	The same requirements, with the simulation environment, starter code, and API guidance; agents select and interpret their own tests.
Full harness	Simulation support, independent requirement-level test binding, and diagnostic execution feedback throughout revision.
Binary feedback	The same harness workflow, with pass/fail simulation feedback in place of diagnostic details.

Requirements-only setting. This setting follows ordinary user interaction with an agent: the user describes the desired mechanism and its functions in a natural-language request, and the agent generates a corresponding CAD assembly. The agent chooses how to construct the mechanism and verify its functionality, without requirement-level testbeds or diagnostic feedback from FunCAD-Harness.

B Mechanism Families and Task Taxonomy

B.1 Mechanism Catalog

FunCADBench comprises the 46 mechanism families listed in Table 4, numbered consecutively from 1 to 46. Each family is defined by its required functions and fixed constraints, while allowing different geometries, transmission arrangements, and permitted operating choices. Different designs that satisfy the requirements of a family are treated as alternative realizations rather than additional families. The table summarizes each family's functional requirements and the operations and observations used to evaluate them. The specific components, inputs, measurements, and operating parameters are established through independent binding to each candidate design.

Table 4: The 46 mechanism families in FunCADBench, their functional requirements, and the corresponding operations and observations.

Mechanism	Functional requirements	Operations and observations
1 Hinged toolbox	Open the lid to the target angle against a physical stop, then close it; keep the lid attached and the hinge pin axially retained throughout.	Actuate the handle or lid edge; record lid angle, stop contact, and axial hinge-pin displacement.
2 Rack-and-pinion gate	Raise and lower the gate, including under load, with the required travel per handle revolution while maintaining guidance and respecting travel limits.	Turn the handle; record input rotation, gate travel, load, guide retention, and travel limits.
3 Four-bar wiper	Convert continuous input rotation into repeated sweeping with the required angular range, consistent return, and intact linkage connections.	Rotate the drive shaft; record sweep extrema, return position, cycle count, and accumulated drift.
4 Hand-cranked egg beater	Drive both beaters in opposite directions at the required speed ratio, including under the specified resistance.	Turn the handle; record handle and beater angles and speeds, resisting loads, and shaft retention.
5 Self-locking utility knife	Extend the blade to two positions, retain each position after release, and permit unlocking and full retraction.	Operate the slider; record extension at each position, displacement after release, and the final tip position.
6 Folding drying rack	Unfold both frames and raise the drying surface, stand stably, and fold into the required storage configuration without losing support.	Operate the folding controls; record frame angles, drying-surface height, ground support, and folded dimensions.
7 Bicycle	Provide pedal-driven travel, steering, and bell operation; cargo variants additionally require loaded travel and steering.	Operate the pedals, handlebar, and bell; record drive ratio, wheel contact, stability, and cargo loading where applicable, and assess bell sound separately.
8 Pedal-operated bin	Open and hold the lid while the pedal is pressed, then close the lid and return the pedal after release over repeated cycles.	Press and release the pedal; record lid angle, pedal return, and closure after release.
9 Mechanical alarm clock	Run after winding, display and adjust time, retain an independently set alarm, and trigger and stop the alarm without interrupting timekeeping.	Operate winding, time-setting, alarm-setting, and stop controls; record hand motion, settings, and striking, with sound assessed through a separate physical test.
10 Kitchen tongs	Grasp an object, retain it while the tongs are lifted, and reopen on release to return the object to the table.	Squeeze and lift the tongs; record jaw gap, object height and slip, and table contact.
11 Retractable ballpoint pen	Complete two extension–retraction cycles, with successive button presses and releases alternating and retaining the tip state.	Press and release the button; record press count, released button position, and tip state.
12 Nail clipper	Repeatedly close the cutting edges to the required gap and reopen them when the lever is released.	Operate the lever; record edge separation, reopening, and pressing force; material cutting requires a separate physical test.
13 One-way ratchet screwdriver	Accumulate forward output rotation under reciprocating handle motion, both unloaded and under the specified resisting torque.	Turn the handle back and forth; record net output rotation, reverse motion during return, and resisting torque.
14 Linearly actuated scissor lift	Raise and lower the platform within the input-force limit while maintaining its orientation, guidance, and support, and hold its height under load.	Operate the lift input; record platform height and tilt, applied load, driving force, and holding drift.

Continued on next page

Table 4 continued

Mechanism	Functional requirements	Operations and observations
15 Equal-arm balance	Balance equal loads, tilt toward the heavier side, recover after removing excess load, retain freely suspended pans, and indicate mass difference.	Load and unload the pans; record mass difference, beam tilt, pan pose, pointer reading, and settling behavior.
16 Toggle clamp / press	Close and reopen using the selected variant's controls; self-locking variants must resist an opening load, while pressing variants must repeat loaded strokes and return.	Operate the handle or release control; record closing travel, clamping or pressing force, loaded drift, and return.
17 Dry piston-pump mechanism	Repeat piston strokes with the required travel and return position, both unloaded and under resistance, while maintaining guidance and connections.	Rotate the crank; record piston position, stroke, period, and resistance; the task evaluates dry motion rather than fluid pumping.
18 Three-digit combination lock	Adjust each wheel independently, unlock with the correct code, re-lock after reinsertion and scrambling, and reject a single wrong digit at each position.	Operate the wheels and bolt; record code settings, locking-bar or bolt travel, and blocking under incorrect codes.
19 Two-digit mechanical counter	Count correctly through $09 \rightarrow 10$ and $19 \rightarrow 20$, increment once per forward stroke, prevent reverse counting on return, and retain the reading at rest.	Operate the push rod; record the units and tens readings after each forward stroke, return stroke, and release.
20 Three-ball dispenser	Dispense exactly three balls in each of three cycles, reaching cumulative totals of 3, 6, and 9; refill between cycles while retaining unused balls and conserving the total count.	Operate the dispensing slider; track individual balls, counts, and cycle timing across the hopper, metering chamber, and receiving area.
21 Gravity timer	Run autonomously after release, trigger within the specified time window without triggering early, retain a visible indicator, and repeat after reset.	Operate reset and release; record weight position, trigger time, and indicator retention.
22 Slider-crank push rod	Complete three unloaded reciprocating cycles with the required forward and return travel, then repeat under the specified load.	Rotate the crank; record slider position, cycle count, stroke length, and applied load.
23 Gravity-pawl turntable	Allow forward rotation with gravity-driven pawl re-engagement, block reverse rotation under the specified torque, and resume forward rotation.	Rotate the turntable; record forward and reverse angles, pawl seating position and time, and reverse torque.
24 Sliding-bolt gate	Remain locked under pushing with the bolt inserted, then open to the required angle after withdrawal while retaining hinge support and bolt guidance.	Operate the bolt and gate handle; record bolt travel, opening angle, and gate motion while locked.
25 Three-stage gear reducer	Transmit rotation at 3:1 per stage and 27:1 overall, operate under three braking loads, and decelerate naturally after input removal.	Drive only the input shaft; record each shaft's angle and speed, input energy, and braking dissipation.
26 Desktop display turntable	Rotate to $90^\circ \pm 3^\circ$ and return within $\pm 3^\circ$ of the initial position without eccentric motion or axial separation.	Apply bounded force to the manual tab; record rotation angle, return error, eccentricity, and axial displacement.
27 Small swing gate	Open to $60^\circ \pm 3^\circ$ and return within $\pm 3^\circ$ of the initial position while maintaining a stable frame and preventing hinge-axis slip or disengagement.	Push and pull the gate; record opening angle, return error, and downward displacement along the hinge axis.
28 Tilting desktop mirror	Tilt $30^\circ \pm 3^\circ$ from upright and return within $\pm 3^\circ$ of the initial position while maintaining base stability and pivot support.	Rotate the mirror frame; record tilt and axial displacement; angle locking is not required.
29 Folding wall hook	Deploy the hook through $90^\circ \pm 3^\circ$ from its stowed position and return within $\pm 3^\circ$ while retaining the fixed base and hinge connection.	Operate the hook arm; record rotation angle, return error, and axial hinge displacement.

Continued on next page

Table 4 continued

Mechanism	Functional requirements	Operations and observations
30 Spring clothespin	Open when squeezed, retain the specified test piece after release, release it when squeezed again, and repeatedly return to the closed position.	Squeeze and release the tails; record jaw gap, test-piece slip, release, and the specified load.
31 Scissors	Repeatedly open and close through the required angular range while retaining the pivot and avoiding blade interference.	Operate the handles; record blade angle, opening, and pivot retention; opening and closing do not establish material-cutting capability.
32 Ratchet hand winch	Wind under load, hold during handle return or reverse drum torque, permit controlled payout, and re-engage to hold and wind again.	Operate the handle and release control; record drum rotation, load displacement, and drift after stopping.
33 Spring-latch door lock	Retract the latch when the handle turns, extend it on release, and automatically retract and re-engage against the strike during door closure.	Operate the handle and close the door; record latch travel, door position, and contact with or entry into the strike.
34 Reversible dual-pawl ratchet wrench	Drive the head in the selected direction without reversing it on the return stroke; switch direction and repeat, including under resistance.	Operate the handle and selector; record output advance, reverse motion on return, and resisting torque in both modes.
35 Full-cycle ratchet crimper	Reach the required loaded stroke and force, prevent opening after a partial stroke, and release for full reopening at the endpoint over repeated cycles.	Operate the handles; record head travel, force, and pawl state; metal forming and electrical-joint quality are outside this check.
36 Geneva indexing table with overload clutch	Index by 90° with stable dwell under load, resist dwell disturbances, slip under obstruction, and resume correct indexing after the obstruction is removed.	Drive the input and apply disturbances or obstruction; record table angle, dwell drift, rotations on both sides of the clutch, and load.
37 Folding shopping cart	Carry the specified load while rolling and permit handle folding and unfolding without losing assembly integrity.	Operate the pulling and handle-folding controls; record travel, load, frame support, wheel motion, and handle angle.
38 Mechanical 4×4 puzzle cube	Rotate and align outer and inner layers separately without unintended motion of other layers or piece loss, and execute successive turns about different axes.	Turn the selected layers; record inner and outer layer angles, unintended layer motion, and piece retention.
39 Twin-cylinder engine linkage	Drive both pistons through repeated strokes with the required phase relationship and travel while maintaining guidance and avoiding interference.	Rotate the crankshaft; record piston positions, phase, and stroke; the task evaluates transmission rather than combustion.
40 Four-station indexing table	Cycle through the designated stations under continuous input, alternating motion and dwell without disengagement or jamming.	Rotate the input shaft; record table angle, stopping positions, and dwell duration.
41 Synchronized parallel gripper	Open and close symmetrically from a single input while maintaining parallel jaw faces and guidance, and grasp and retain a test piece under load; release and repeat the grasp–hold–release sequence.	Operate the gripper; record jaw travel, face angle, gripping force, test-piece slip, and the state left for the next grasp.
42 Releasable reduction winch	Lift under load through a reduction transmission, hold against reverse drum torque after input stops, permit controlled payout, and re-engage for holding and winding.	Operate the handle and release control; record input and drum rotation, load displacement, holding drift, and release and re-locking behavior.

Continued on next page

Table 4 continued

Mechanism	Functional requirements	Operations and observations
43 Baby stroller	Roll and steer under load, engage and release the brake, and, for folding variants, fold, unfold, and re-lock the frame.	Operate the push handle, steering, brake, and folding controls; record body pose, wheel motion, load support, and residual displacement after braking.
44 Height- and tilt-adjustable worktable	Reach the required height and tilt ranges, hold both settings under load, and permit unlocking, readjustment, and re-locking.	Operate the lift, tilt, and locking controls; record height, angle, load, and drift after locking.
45 Coin sorter and counter	Sort a continuous mixed coin stream to the correct outlets, count each coin exactly once, and resume after pauses or collection-bin emptying without count errors.	Operate the feeder; record coin type and size, outlet assignment, outlet counts, counter readings, and jams.
46 Interchangeable-tune music box	Play and stop in the required direction and rhythm, actuate sound elements in sequence with recovery, and accept a replacement tune component.	Operate power or winding, stop, and tune replacement; record score position and actuation sequence, with melody recognition assessed through a separate physical test.

B.2 Taxonomy Dimensions and Representative Mechanisms

We characterize the mechanical demands of the tasks along five complementary dimensions: motion primitives, topology, complexity, contact, and geometric difficulty. These dimensions are not mutually exclusive categories: a mechanism family typically involves several dimensions, and task variants may place different demands on each.

Motion primitives. This dimension describes how a mechanism converts and transmits motion, including input–output direction, phase relationships, and repeated or intermittent operation. In the four-bar wiper (3), continuous crank rotation must produce repeated wiper oscillation, so evaluation must capture the input–output relationship over complete motion cycles.

Topology. This dimension describes connections, branches, and closed kinematic chains, together with the compatibility of the constraints they impose. In the scissor lift (14), revolute and prismatic connections form closed loops that constrain platform motion; the connections must therefore be individually valid and collectively kinematically compatible.

Complexity. This dimension describes interactions among components and subsystems, state transitions, required operating sequences, and part count. The mechanical alarm clock (9) couples energy supply, timekeeping, and alarm operation through winding, running, and triggering, requiring coordination across subsystems and states.

Contact. This dimension describes how engagement, sliding, locking, friction, and changes in contact state enable or prevent a behavior. In the self-locking utility knife (5), pressing the release changes the contact state to permit sliding, after which engagement at another detent restores locking.

Geometric difficulty. This dimension describes the feature construction, alignment, clearance control, and assembly access needed to realize the intended mechanical relationships. The reversible dual-pawl ratchet wrench (34) illustrates these demands through its pawl–tooth geometry, pivot fit, and local clearances, whose shapes and relative placement must support the intended engagement.

The five dimensions interact: adding components may introduce new contacts, while changing the topology may simplify geometry but complicate operation. The taxonomy therefore describes the mechanical diversity of the benchmark, while performance on an individual family generally reflects the combined effects of several capabilities.

C Functional Specifications and Diagnostic Cases

We use Fable's design trajectory for the synchronized parallel gripper (Task 41) to illustrate how functional requirements guide construction, execution, and revision, and how the resulting evidence establishes completion. Appendix C.6 adds two repair cases that connect functional feedback to local geometric changes and renewed verification.

C.1 From Input Requirements to Executable Tests

Table 5 connects three levels of specification for the cases below. The *input requirement* states the intended behavior. The *functional test template* decomposes that behavior into operations, observations, and acceptance checks, with interfaces and parameters to be bound to a candidate. The *candidate-specific simulation test* supplies the actual controls, environment, measurements, and declared parameter values. Registered tasks reuse the template across candidates; open-ended requests require a new template and independent review, as described in Appendix A.

Table 5: From input requirements to functional test templates and recorded candidate-specific simulation tests. Prompt clauses are condensed; numerical settings refer to the named candidate and revision.

Input requirement	Functional test template	Candidate-specific simulation test
Task 41: synchronized parallel gripper — Fable, r02		
A single input opens and closes both jaws with symmetric motion and parallel gripping faces.	*Motion and guidance.* Operate the normal input; check each jaw's travel, synchronization, face parallelism, guide deviation, and available opening.	Apply force to the central plunger along its sliding axis, capped at 35 N. Measure the actual jaws: each travels at least 6 mm; synchronization and guide errors stay within 0.5 mm, parallelism within 0.01 rad, and the released opening reaches at least 50 mm.
Hold the test piece without slipping or falling under the specified load.	*Loaded holding.* Apply a load to the grasped piece; verify actual load, minimum jaw–piece contact force, and three-dimensional slip relative to the jaws.	Apply 5 N tangentially to the free test block during 1–3 and 5–7 s. Require at least 10 N grip force, measured mean load of at least 4.5 N, and slip of at most 1 mm in each loaded phase.
Grasp, release, and repeat using the same operating input.	*Repeated use.* Execute two close–hold–release cycles in one trial, retaining the workpiece state between cycles. Apply the same checks to both cycles.	The `repeated_loaded_grip` trial lasts 8 s. Plunger targets are +40 mm for closing and 0 mm for opening; physical contact limits actual travel. The second cycle begins at 4 s without resetting the block. All 16 checks are evaluated on this continuous trial.
Task 33: spring-return door latch — Grok, r06		
Turning the handle retracts the latch so the door can open.	*Unlocking and opening.* Retract the latch through handle operation before opening the door; observe latch protrusion and door position.	Bind inputs to the actual handle and door hinges, with torque caps of 3 and 12 N m. Target a handle angle of -70° and a door opening of 0.4 rad. Require latch protrusion of at most 8 mm and door-position error of at most 0.12 rad.
Releasing the handle lets the latch extend automatically.	*Passive return.* Remove handle actuation while the door is open; check that the spring restores latch extension.	After opening, stop driving the handle. Read the actual nose position relative to the door and require at least 10 mm protrusion at the end of the return phase.
Closing compresses the beveled latch, which springs into the strike plate and locks again; repeat the sequence.	*Contact-mediated re-locking.* Close the door; check strike contact, temporary latch retraction, closed position, and engagement after door actuation ends. Repeat the full sequence.	The `repeated_open_close` trial uses 0.8 s phases and two cycles over 8.8 s. Measure strike-contact force and exported latch-nose penetration into the actual strike opening. Require contact, protrusion of at most 8 mm during closure, and positive final engagement in both cycles.

Continued on next page

Table 5 continued

Input requirement	Functional test template	Candidate-specific simulation test
Task 39: twin-cylinder linkage — Grok, r02		
Continuous crankshaft rotation drives both pistons repeatedly with the design's strokes and phase relationship.	*Transmission and phase.* Drive the crank; measure both piston trajectories, complete reciprocation counts, and crank angles at matched piston extrema.	The `continuous_crank_drive` trial drives the shaft for 3.5 s at a target of 2π rad/s with a 1 N m torque cap. Require at least two cycles per piston, 24 mm stroke within 2 mm, and phase separation of π within 0.1 rad over at least two matched extrema pairs.
Both pistons stay within their guides without unintended interference or jamming.	*Guidance and interference.* Monitor guide deviation and unintended contact throughout the same crank-driven trial; distinguish normal bearing contacts from prohibited collisions.	Track both ends of each piston skirt in the frame coordinates; require guide error of at most 1 mm. Sum contact forces outside the identified bearing regions and require at most 1 N. Together with the motion and phase checks, the trial evaluates eight functional checks.

The mapping is not one requirement per simulation run: several checks can share a continuous trial, preserving the states needed for later operations. For example, the second grasp uses the block left by the first release, and re-locking is checked after closing inputs have been removed. The trials also undergo source-geometry auditing at sampled poses, which is separate from the functional checks summarized above (Appendix D). These protocols test the required behaviors for particular designs; a different mechanism may need a different input location, operating policy, or observation mapping.

C.2 Worked Example: Designing and Refining a Parallel Gripper

The task requires a single input to drive two synchronized, parallel jaws that grasp a workpiece, hold it under load, release it, and repeat the operation. Fig. 9 follows the agent from choosing a linkage and inspecting its CAD assembly to developing tests, interpreting formal feedback, and verifying the revised design.

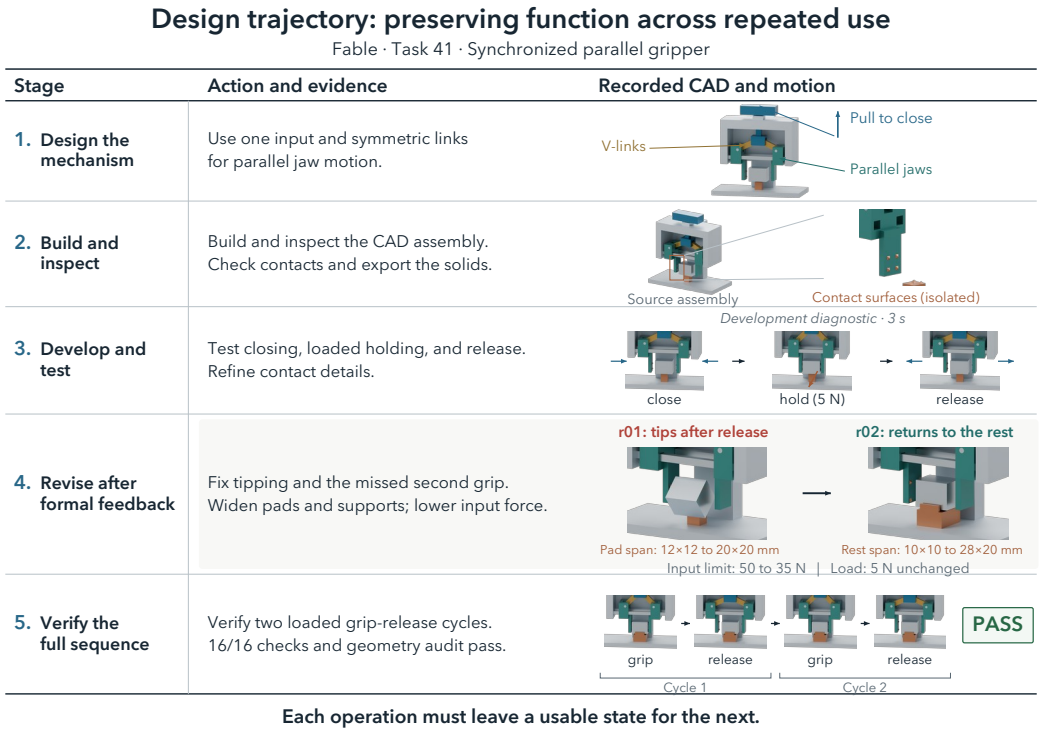


Figure 9: Fable's design trajectory for the synchronized parallel gripper (Task 41). Construction and development tests establish the initial grasping behavior; formal feedback exposes a failure during repeated use. The agent revises the contact geometry, workpiece support, and permitted input effort before the final candidate passes both loaded grasp–release cycles and the source-geometry audit.

The first submitted design completes the initial loaded grasp, but its revision report identifies a subsequent failure: after release, the workpiece tips off its rest, leaving the jaws to close on empty space in the second cycle. The agent widens the contact pads and support footprint and lowers the permitted input effort, while retaining the external test load. The revised candidate passes the complete sequence. This trajectory illustrates why a successful operation must leave the mechanism and its environment in a usable state for the next operation.

C.3 Fixed Requirements, Adjustable Parameters, and Binding

The specification distinguishes *functional requirements*, *fixed constraints*, and *adjustable operating parameters*. For Task 41, synchronized parallel motion, loaded holding, release, and repeated use remain required throughout revision. The single-input constraint also remains fixed. As a registered benchmark task, it uses a frozen test template; *candidate-specific binding* makes that template executable for the submitted mechanism.

Candidate-specific binding. The final evaluation agent connects the template to the candidate's *mechanism*, *environment*, *control policy*, *observations*, and *pass criteria*. The mechanism uses a central plunger, symmetric links, and guided jaws to grasp a free workpiece resting on a support; an external load is applied during holding. The policy pulls the plunger to close the jaws, holds under load, releases, and repeats. Observations include jaw synchronization and parallelism, grip force, workpiece slip, and its state after release. The pass criteria require these measurements to satisfy the task's motion and holding requirements throughout the sequence.

Operating protocol. Different gripper designs may place the input, contact surfaces, and supports differently, so the locations and directions of actuation, load application, and measurement must follow the candidate. The constructor's operating instructions inform these choices, while the test-binding agent checks them against the delivered geometry and mechanics. Binding changes how the required behavior is exercised and observed; it preserves the behaviors that must be established.

Parameter provenance and permitted adjustments. Each parameter retains its value and supporting basis, whether derived from the task, source documentation, geometric measurements, or an explicit constructor decision. A permitted adjustment records the original target, revised target, and reason for the change, following Appendix A.4. In this trajectory, the input-effort limit changes from 50 to 35 N, while the external 5 N load and the requirements for loaded holding, release, and repeated use remain unchanged. The final candidate is evaluated with the 35 N input-effort limit, while the external 5 N load and the requirements for loaded holding, release, and repeated use remain unchanged.

C.4 State Dependencies and Completion

Each trial starts from a fresh instance of the delivered configuration, while successive operating phases preserve the evolving physical state. In the gripper test, opening the jaws releases the workpiece without resetting its position before the next grasp. An isolated successful grasp therefore does not establish repeatability: the released workpiece must remain accessible to the mechanism.

Whole-task completion requires every required function and the applicable physical-validity checks to pass under the declared operating conditions and modeling assumptions. For the final gripper, this includes both loaded grasp–release cycles and the source-geometry audit. All evidence must refer to the same delivered configuration, which remains unchanged during evaluation. After a revision, the binding and evaluation must be checked against the updated candidate; successful checks from different designs cannot be combined to establish completion of one design.

C.5 Diagnostic Interpretation

The gripper trajectory illustrates how feedback separates verified behavior, observed failures, and missing evidence.

Partial functional evidence. Development tests of closing, loaded holding, and release provide useful evidence for refining contacts, but do not by themselves establish that the complete operating sequence succeeds. Likewise, a passed first grasp leaves the second grasp unverified.

Observed functional failure. In the first submission, the second closure fails to grip the workpiece. The failed grip and holding checks identify the unmet behavior; examining the preceding release explains why that behavior is no longer possible from the resulting state. Repair therefore involves the workpiece support and release behavior, in addition to the geometry that produces the grasp.

Physical-validity failure. During development of the revision, a flat open-stop contact triggers a source-geometry overlap check and is replaced with three small support buttons. A desired motion can therefore coexist with a geometric violation. Functional measurements and source-geometry findings provide distinct feedback and both contribute to the completion decision.

Insufficient evidence. If a required operation is not executed or its necessary observation is unavailable, the corresponding function remains inconclusive. For example, omitting the second loaded grasp would leave repeatability unestablished even if the first cycle passed. Missing evidence establishes neither mechanical failure nor completion.

Test outcomes and defect attribution. A failed grip check identifies an unmet criterion, but does not by itself distinguish inadequate contact force from a misplaced workpiece. That distinction requires evidence about contacts and the preceding motion; here, the revision report links the missed second grasp to post-release tipping. When assessing diagnostic correctness, reference defect labels must be established independently of the test outcome being evaluated.

C.6 Repair Trajectories

The following cases show how functional feedback and geometry checks guide successive revisions of an existing design.

Restoring re-locking after closure (Task 33). Fig. 8 follows Grok's repair of a spring-return door latch. In revision r04, the door opens successfully, but the latch rests on the strike-plate edge after closing instead of entering its opening. Both re-locking checks fail, leaving 13 of 15 functional checks passed. Grok reshapes the entry bevel and adds a catching shoulder; revision r05 restores engagement after both closures and passes all 15 functional checks. A remaining latch–faceplate interference is then removed through a local faceplate relief in r06, which also passes the source-geometry audit. Test parameters remain unchanged throughout these revisions. The case distinguishes reaching a closed door position from restoring the latch engagement required for re-locking, and separates functional recovery from subsequent geometric validation.

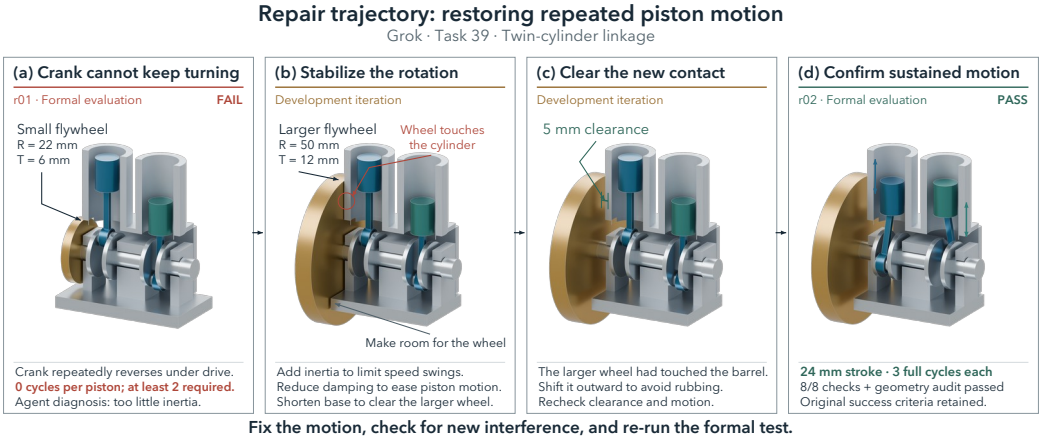


Figure 10: Grok repairs the twin-cylinder linkage (Task 39): enlarge the flywheel, clear the resulting cylinder contact, and verify repeated piston motion.

Restoring sustained piston motion (Task 39). Fig. 10 follows Grok's repair of a twin-cylinder linkage. In revision r01, the crank repeatedly reverses under drive and neither piston completes a full cycle. The agent diagnoses insufficient rotational inertia and enlarges the flywheel, while reducing guide damping and shortening the bedplate. Intermediate development checks then reveal interference between the larger flywheel and a cylinder barrel. Moving the wheel outward provides approximately 5 mm of clearance. In the formal evaluation of r02, both pistons complete three cycles with approximately 24 mm stroke; all eight functional checks and the source-geometry audit pass under the original success criteria. This case shows how a repair aimed at sustained motion can introduce a new geometric conflict, requiring clearance checks followed by verification of the complete behavior.

D Scene Interface and Physical Execution Details

D.1 Scene Interface and Initialization

The physical scene associates the submitted geometry with rigid bodies, connections, material properties, external supports, and gravity. Source preparation constructs the simulation model from these geometric and physical declarations before test binding. The binding connects the template's operating inputs and observation signals to the prepared model. Prechecks verify that these interfaces are available and that the source bodies and their physical-update routines are retained.

During execution, prescribed inputs are applied at the bound operating interfaces, subject to the force or torque limits specified by the test protocol. Trial initialization and state continuity follow the rules in Appendix C.4. Parameter provenance and permitted adjustments are recorded as described in Appendix C.3.

D.2 Contact, Connections, and Equivalent Models

The registered-template execution path uses the NSC system in Project Chrono (Tasora et al., 2016) for rigid-body dynamics and Bullet for collision detection. Collision representations are constructed from the retained source geometry, and the material and contact parameters used during execution are recorded with the run.

The supported connection types are revolute, prismatic, spherical, and fixed. Each connection is associated with finite regions on the source geometry and remains active only while those regions overlap. If the regions separate, the connection is disabled before the next integration step and remains released for the rest of the trial.

Selected mechanical effects, including spring storage, elastic response, friction-clutch behavior, detents, and gear engagement, can be represented by explicitly declared equivalent models. These models apply force or torque laws to the corresponding source components while the native solver computes the resulting rigid-body motion. Their parameters, source correspondence, and operating ranges are specified before execution.

The original CAD geometry is retained, and any exclusions from motion-geometry auditing are restricted to declared elastic reference bodies or explicitly bounded local regions. Other geometry and contacts remain subject to the applicable checks.

D.3 Trajectory Recording and Geometry Auditing

The primary registered-template path records measurement signals at every native integration step. These signals include the motion, connection states, and contact information required by the bound tests. Body poses are saved at nominal 20 ms intervals, with additional poses retained at operating-phase boundaries.

Functional assertions are evaluated from the recorded measurement signals, while source-geometry auditing uses the saved body poses. The audit selects poses at 10 Hz, includes the initial and final poses, and checks the original source geometry for violations of the applicable geometric criteria. The 20 ms recording interval determines the retained motion detail, whereas the 10 Hz audit frequency

Table 6: Task 25 reducer timings (1 ms step; standard test budget: 300 s). Different designs and simulated durations preclude a controlled speedup comparison.

Interaction model	Wall/step (s)	Sim. time (s)	Total wall (s)
Native tooth contact (light)	0.894	6.0	**5,364**
Equivalent spur gears (light/medium/heavy)	0.113–0.117	1.2	**136–141**

Table 7: Generation-agent and final-evaluation configurations

Role	**Model identifier**	**Wrapper**	**Requested reasoning setting**
Constructor	`claude-fable-5-1-thinking-high`	Cursor	High
Constructor	`cursor-grok-4.6-xhigh`	Cursor	Extra high
Constructor	`kimi-code/k3`	Kimi CLI	High*
Constructor	`gpt-6-astra`	Codex	High
Final evaluation agent	`gpt-6-astra`	Codex	Medium

*For Kimi CLI, High records the requested setting; the integration does not expose a separate reasoning-effort control.

determines when the geometry is inspected. Geometric conclusions apply to the inspected poses within the declared modeling scope.

If a trial fails or is interrupted, the measurements and poses recorded up to that point are retained together with its execution status.

E Agent Configurations, Budgets, and Evaluation Protocol

E.1 Model and Agent Configurations

Table 7 lists the model identifiers, agent wrappers, and requested reasoning settings used during generation and final evaluation. During generation, each agent uses its own model configuration. Final batch evaluation uses GPT-6 Astra with a one-million-token context allowance and medium reasoning across all experimental settings.

Both Fable 5.1 and GPT-6 Astra are configured with a context allowance of one million tokens. Each constructor receives the registered English task prompt and shared CAD-delivery instructions, with tool access and feedback determined by the experimental condition described in Appendix A.

E.2 Attempts and Time Accounting

An independent attempt includes construction, testing, and revision for one mechanism family within a three-hour active-time budget. In the single-attempt settings, each family receives one such attempt; internal tests, retries, and revisions do not count as additional independent attempts.

Active time is computed from the union of observed working intervals, so overlapping activity is counted only once. For intervals $I_1, \ldots, I_m$, we define

$$T_{\text{active}} = \text{duration}\left(\bigcup_{j=1}^{m} I_j\right). \tag{1}$$

Each formal native simulation test has a wall-clock limit of 300 seconds, with simulated duration recorded separately.

E.3 Artifact Selection and Outcome Aggregation

Acceptance is determined from the final delivered configuration under its final declared operating target, following the completion rules in Appendix C.4. The record retains the original target, any

adjustments, and the conditions actually evaluated. Each reported outcome is linked to the evaluated design revision, its test protocol, and the corresponding execution evidence.

Each attempt is assigned to one of three mutually exclusive outcome groups. *First-submission pass* denotes completion on the first candidate submitted to the formal harness. *Pass after revision* denotes completion on a subsequent candidate after revision. *Not passed* denotes the absence of recorded completion by the end of the attempt, including cases with unfinished evaluation.

Submission order is established from program events and preparation snapshots rather than revision identifiers alone. The first formal submission may follow substantial development and self-testing, so first-submission success does not imply success on the first generated design.

The three outcome counts sum to the number of available attempts for each configuration, and total completion is the sum of first-submission and post-revision passes. For the single-attempt configurations, Fig. 6 reports median active time across all 46 tasks, including those without recorded success.

Token use is the median of cumulative constructor input and output tokens per task, in millions, over tasks with recorded usage. Estimated API cost is computed per task from uncached input, cache reads, cache writes, and output, then summarized by the median. We use official standard API prices checked on September 25, 2026, with short-context rates for Astra and Grok and 5-minute cache writes for Fable. The exported counters do not resolve per-request context tiers or cache lifetimes, so long-context surcharges and 1-hour cache-write rates are not included. These estimates cover recorded constructor usage only, excluding reviewer usage and external tool charges; unreported turns can leave a task's cost incomplete.

The Grok + history setting allows up to eight sequential attempts per task on the same 46-task cohort, with access to earlier records. Its generated machines undergo the same final batch evaluation as the other settings.

E.4 Task-Level Completion Results

Table 8 lists the task-level outcomes for the four single-attempt full-harness configurations underlying Fig. 6. Task IDs and names follow the mechanism catalog in Appendix B.

Table 8: Task-level recorded completion in the four single-attempt full-harness configurations. A check mark denotes completion under formal evaluation; a dash denotes no recorded completion, including unfinished evaluation. Each configuration has one independent attempt per task.

ID	Mechanism	Astra	Kimi	Fable	Grok
1	Hinged toolbox	—	—	—	—
2	Rack-and-pinion gate	✓	✓	✓	✓
3	Four-bar wiper	✓	✓	✓	✓
4	Hand-cranked egg beater	✓	✓	✓	✓
5	Self-locking utility knife	✓	—	✓	✓
6	Folding drying rack	—	—	✓	—
7	Bicycle	—	—	—	—
8	Pedal-operated bin	✓	—	✓	✓
9	Mechanical alarm clock	—	—	—	—
10	Kitchen tongs	—	—	✓	—
11	Retractable ballpoint pen	—	—	✓	—
12	Nail clipper	✓	—	✓	—
13	One-way ratchet screwdriver	—	—	✓	—
14	Linearly actuated scissor lift	—	—	✓	—
15	Equal-arm balance	✓	✓	✓	✓
16	Toggle clamp / press	—	✓	✓	✓
17	Dry piston-pump mechanism	✓	✓	✓	✓
18	Three-digit combination lock	✓	—	✓	—
19	Two-digit mechanical counter	✓	—	✓	—
20	Three-ball dispenser	—	—	—	—
21	Gravity timer	✓	—	✓	✓
22	Slider-crank push rod	✓	✓	✓	✓
23	Gravity-pawl turntable	✓	—	—	—
24	Sliding-bolt gate	✓	✓	✓	✓
25	Three-stage gear reducer	—	—	✓	—
26	Desktop display turntable	✓	✓	✓	✓
27	Small swing gate	✓	✓	✓	✓
28	Tilting desktop mirror	✓	✓	✓	✓
29	Folding wall hook	✓	✓	✓	✓
30	Spring clothespin	—	—	✓	✓
31	Scissors	✓	✓	✓	✓
32	Ratchet hand winch	—	—	—	—
33	Spring-latch door lock	✓	—	—	✓
34	Reversible dual-pawl ratchet wrench	—	—	—	—
35	Full-cycle ratchet crimper	—	—	✓	—
36	Geneva indexing table with overload clutch	—	—	—	—
37	Folding shopping cart	✓	—	✓	—
38	Mechanical 4×4 puzzle cube	—	—	—	—
39	Twin-cylinder engine linkage	✓	✓	✓	✓
40	Four-station indexing table	—	—	✓	—
41	Synchronized parallel gripper	—	—	✓	—
42	Releasable reduction winch	—	—	—	—
43	Baby stroller	—	—	—	—
44	Height- and tilt-adjustable worktable	—	—	✓	—
45	Coin sorter and counter	—	—	—	—
46	Interchangeable-tune music box	—	—	—	—
Total completed		**22/46**	**14/46**	**32/46**	**19/46**

E.5 Demand-Weighted Completion and Complexity Strata

The radar in Fig. 6 uses five requirement-based demand grades derived from the taxonomy. Motion coordination grades motion primitives; structural coupling grades topology; functional integration grades coordination among functions and states; contact coordination grades engagement and contact-state requirements; and geometric constraints grade fits, swept clearances, and interacting geometric relations. A grade of 1 denotes a local or direct requirement, 2 denotes a coupled relation, and 3 denotes multiple interacting relations or operating states, as specified below. These are descriptive ordinal annotations, rather than calibrated difficulty units or independently measured cognitive abilities. The functional-integration grade and the original low/medium/high complexity label are related annotations but are not interchangeable.

For each axis, weighted completion is the sum of that axis's grades on completed tasks divided by the sum on all 46 tasks. Using grades 1, 2, and 3 as weights is a linear weighting convention; every task contributes positively to every denominator. The same weights are used for all models.

Table 9: Anchors for the three mechanical-demand grades.

Axis	Grade 1	Grade 2	Grade 3
Motion	Direct operation of one degree of freedom	One input–output relation	Multiple stages or coordinated outputs
Structure	Local support or connection	One jointly constrained chain	Multiple constraint groups sharing parts or state
Function	One basic operation and return	Coupled functions or operating conditions	Interdependent functions or conditional states
Contact	Continuous support or transmission	Engagement/release or force-dependent function	Several contacts coordinated across states
Geometry	Local fit, alignment, or retention	Swept clearance or one meshing relation	Interacting clearances, retention, or meshing constraints

Table 10: Task demand grades used in the weighted-completion analysis. Columns M, S, F, C, and G denote motion, structure, function, contact, and geometry. Task IDs follow the mechanism catalog.

Task	M	S	F	C	G	Task	M	S	F	C	G
1	1	1	1	1	1	24	2	2	2	2	2
2	2	2	2	1	2	25	3	2	2	1	3
3	2	2	1	1	2	26	1	1	1	1	1
4	3	2	2	1	2	27	1	1	1	1	1
5	1	2	3	3	2	28	1	1	1	1	1
6	3	3	2	2	3	29	1	1	1	1	1
7	3	3	3	2	3	30	2	2	2	2	1
8	2	2	2	1	2	31	1	1	1	1	2
9	3	3	3	3	3	32	2	3	3	3	2
10	2	2	2	2	1	33	2	3	3	3	3
11	2	2	3	3	2	34	3	3	3	3	3
12	2	2	2	2	2	35	3	3	3	3	3
13	2	2	2	2	2	36	3	3	3	3	3
14	2	2	2	2	2	37	2	2	2	1	2
15	2	2	2	1	1	38	3	3	3	2	3
16	2	2	2	2	2	39	3	3	1	1	3
17	2	2	2	1	2	40	2	2	2	2	2
18	3	3	3	3	3	41	3	2	2	2	2
19	3	3	3	3	3	42	2	3	3	3	2
20	2	3	3	3	3	43	3	3	2	2	2
21	2	2	3	2	2	44	3	3	3	2	2
22	2	2	2	1	2	45	3	3	3	3	3
23	2	2	2	2	2	46	3	3	3	3	3

Table 11: Recorded completion within the benchmark's original complexity classes. These classes were not constructed by binning the radar scores.

Complexity	Tasks	Astra	Kimi	Fable	Grok
Low	11	9/11	8/11	9/11	9/11
Medium	20	12/20	6/20	18/20	10/20
High	15	1/15	0/15	5/15	0/15



Figure 11: Agent actions and tool execution time on 41 matched tasks. (a) Constructor tool-call shares, excluding records with missing arguments; mixed and unresolved calls are grouped as Others. (b) Observed standard-tool execution time, combining simulation and cost probes; Concurrent denotes overlapping execution across categories. Shares are normalized within each task and then averaged with equal task weights. Annotation and timing details appear in Appendix E.6.

E.6 Tool-Use Annotation and Execution-Time Profiles

Tool-call classification. Each deduplicated constructor tool call receives one primary-purpose label based on its arguments, outputs, and any recoverable helper-script content from preceding events. *Information access* covers reading task requirements, tool documentation, and source code. *Design, build, and test implementation* covers source edits, artifact construction, and test writing. *Geometry checks* include explicit validation calls, custom geometric measurements, and motion-geometry audits. *Simulation* includes development runs, cost probes, and custom dynamics trials. *Visual evidence* includes preview generation and image-access operations, while *feedback analysis* covers reading test results and analyzing recorded traces. *Execution and workflow management* includes environment setup, file transfer, process monitoring, and artifact bookkeeping. Mixed-purpose calls and calls whose purpose cannot be resolved are assigned to *Others*.

These labels describe observed actions rather than their success or the model's underlying reasoning ability. In particular, an image-access operation does not establish that the model successfully inspected or understood the image.

Sample selection and normalization. The analysis uses 46 task families with observed standard-tool execution time for all four configurations. Calls with missing or truncated arguments that cannot be recovered are excluded before normalization. For this matched sample, 1,883 Kimi calls and one Grok call are excluded, leaving 6,035, 6,018, 8,474, and 22,223 calls for Astra, Kimi, Fable, and Grok, respectively.

Category shares are first computed within each task and then averaged with equal weight across tasks. This procedure prevents tasks with unusually many tool calls from dominating the aggregate profile.

Execution-time accounting. Execution-time shares are computed from independently recorded start and end times for standard build, check, simulate, probe, and preview operations. Overlapping intervals within the same category are counted once, while intervals containing simultaneous activity

from multiple categories are assigned to *Concurrent*. Intervals with missing endpoints or unresolved zero duration are excluded.

For each task, category durations are normalized by the total observed standard-tool execution time and then averaged equally across the same 41 families. These profiles include preparation performed within each tool operation, but do not represent total agent runtime or all categories in the behavioral analysis. Preview time measures rendering, and neither it nor the remaining portion of active time is interpreted as a direct measure of model reasoning.

Timing-coverage check. We additionally examine the 14 task families with complete standard-tool timing records for all four configurations. Simulation remains the largest execution-time category for every configuration, and Grok retains the highest share of build time. These comparisons describe observed execution costs, which depend on the generated mechanisms and execution environments as well as the agents' tool choices.

F Detailed Harness Comparisons

Table 12 retains the success-set overlaps behind Table 2. Independent agents evaluate the final assemblies from each workflow under the task requirements and fixed constraints, selecting simulation strategies and configuring candidate-specific tests as described in Section 5.1.

Table 12: Whole-task successes and their overlap with the full harness. Shared counts tasks solved by both the row's workflow and the full harness for the same model; Only counts successes unique to that workflow relative to the full harness.

	Astra			Grok		
Setting	Passed	Shared	Only	Passed	Shared	Only
Requirements only	1/46	1	0	3/46	3	0
Simulation tools	8/46	7	1	10/46	8	2
Full harness	22/46	—	—	19/46	—	—
Binary feedback	0/46	0	0	8/46	7	1

Relative to simulation tools, the full harness has 15 additional successful tasks for Astra and 11 for Grok; the tools-only setting has one and two successes outside the respective full-harness sets. The net increases are therefore 14 and nine tasks. Relative to simulation tools, both full-harness configurations additionally solve the rack-and-pinion gate, four-bar wiper, and hand-cranked whisk (Tasks 2–4), as well as the self-locking knife and spring-latch lock (Tasks 5 and 33). In the binary-feedback setting, Grok has seven successes shared with the full harness and one additional success.

The comparisons in Section 5.2 examine three roles of the harness: providing access to physical simulation, maintaining coverage of the functional requirements, and returning evidence that guides revision. This appendix specifies the information available in each setting, lists the task-level outcomes, and uses selected trajectories to examine how agents test, revise, and decide when to stop.

F.1 Experimental Settings and Evaluation

Construction-time support. All four settings start from the task's functional requirements and permit CAD construction, source inspection, and static visualization. Table 13 distinguishes the supplied simulation support, responsibility for test coverage, and visible feedback. In the *requirements-only* setting, agents may still write verification code, including their own simulations. The *simulation-tools* setting supplies a simulator and guidance, while leaving the agent to select operating conditions, observations, and acceptance checks. The *full harness* additionally connects the candidate mechanism to requirement-level tests throughout design revision.

Table 13: Support and feedback available during construction.

Setting	Simulation support	Requirement-level testing	Visible evidence
Requirements only	Agent implements any simulation it chooses to use.	Agent chooses its own checks.	Outputs of agent-written checks.
Simulation tools	Simulator, starter code, and tool guidance are supplied.	Agent chooses its own checks.	Agent's simulation outputs and geometric analyses.
Binary feedback	Harness constructs and executes candidate-specific tests.	Required functions remain part of the evaluation workflow.	Pass/fail outcomes; execution and validation diagnostics are hidden.
Full harness	Harness constructs and executes candidate-specific tests.	Required functions remain part of the evaluation workflow.	Failed checks, measured discrepancies, and physical-validity diagnostics.

Scope of binary feedback. The binary-feedback setting restricts both development and formal simulation tests, as well as the standard validation interface, to pass/fail outcomes. The constructor cannot inspect the execution traces, measured residuals, validator diagnostics, or reviewer artifacts underlying these outcomes. It can still inspect its CAD source and static previews, perform its own geometric or analytical calculations, and request additional tests. This comparison asks whether correctness signals alone provide sufficient direction for repair. Simulation tools and binary feedback change both test organization and evidence access; they are complementary comparisons rather than successive levels of a single support variable.

Evaluation and outcome accounting. Requirements-only and simulation-tools designs are adapted for evaluation after construction by a separate agent, whereas the harness settings expose requirement-level evaluation during revision. Post-hoc evaluation includes establishing an executable physical model of the delivered assembly. An incomplete adaptation or execution therefore leaves whole-task completion unestablished, without by itself identifying a mechanical defect. We report the same recorded whole-task outcomes as Table 2; passing an agent-written check is evidence only for that check's scope. The binary-feedback analysis uses the isolated 46-task batch for each model, excluding the earlier development and pilot batches.

F.2 Task-Level Outcomes

Table 14 expands Table 2 by task. Full-harness outcomes reuse the main-experiment runs; each other setting contributes one selected run per model and task.

Table 14: Recorded completion by task and setting. Req.: requirements only; Sim.: simulation tools; Bin.: binary feedback. A check mark denotes recorded whole-task completion; a dash denotes no recorded completion, including incomplete evaluation or partial delivery.

		Astra				Grok			
ID	**Mechanism**	Req.	Sim.	Bin.	Full	Req.	Sim.	Bin.	Full
1	Hinged toolbox	—	—	—	—	—	—	—	—
2	Rack-and-pinion gate	—	—	—	✓	—	—	—	✓
3	Four-bar wiper	—	—	—	✓	—	—	—	✓
4	Hand-cranked egg beater	—	—	—	✓	—	—	—	✓
5	Self-locking utility knife	—	—	—	✓	—	—	—	✓
6	Folding drying rack	—	—	—	—	—	—	—	—
7	Bicycle	—	—	—	—	—	—	—	—
8	Pedal-operated bin	—	✓	—	✓	—	✓	—	✓
9	Mechanical alarm clock	—	—	—	—	—	—	—	—
10	Kitchen tongs	—	—	—	—	—	✓	—	—
11	Retractable ballpoint pen	—	—	—	—	—	—	—	—
12	Nail clipper	—	—	—	✓	—	—	✓	—
13	One-way ratchet screwdriver	—	—	—	—	—	—	—	—
14	Linearly actuated scissor lift	—	—	—	—	—	—	—	—
15	Equal-arm balance	—	—	—	✓	—	—	✓	✓
16	Toggle clamp / press	—	✓	—	—	—	✓	—	✓
17	Dry piston-pump mechanism	—	✓	—	✓	—	✓	—	✓
18	Three-digit combination lock	—	—	—	✓	—	—	—	—
19	Two-digit mechanical counter	—	—	—	✓	—	—	—	—
20	Three-ball dispenser	—	—	—	—	—	—	—	—
21	Gravity timer	—	—	—	✓	—	—	—	✓
22	Slider-crank push rod	—	—	—	✓	—	—	✓	✓
23	Gravity-pawl turntable	—	✓	—	✓	—	✓	—	—
24	Sliding-bolt gate	—	—	—	✓	—	—	✓	✓
25	Three-stage gear reducer	—	—	—	—	—	—	—	—
26	Desktop display turntable	✓	—	—	✓	—	✓	✓	✓
27	Small swing gate	—	✓	—	✓	✓	✓	✓	✓
28	Tilting desktop mirror	—	—	—	✓	✓	—	✓	✓
29	Folding wall hook	—	✓	—	✓	—	✓	✓	✓
30	Spring clothespin	—	—	—	—	—	—	—	✓
31	Scissors	—	✓	—	✓	—	✓	—	✓
32	Ratchet hand winch	—	—	—	—	—	—	—	—
33	Spring-latch door lock	—	—	—	✓	—	—	—	✓
34	Reversible dual-pawl ratchet wrench	—	—	—	—	—	—	—	—
35	Full-cycle ratchet crimper	—	—	—	—	—	—	—	—
36	Geneva indexing table with overload clutch	—	—	—	—	—	—	—	—
37	Folding shopping cart	—	—	—	✓	—	—	—	—
38	Mechanical 4×4 puzzle cube	—	—	—	—	—	—	—	—
39	Twin-cylinder engine linkage	—	✓	—	✓	✓	✓	—	✓
40	Four-station indexing table	—	—	—	—	—	—	—	—
41	Synchronized parallel gripper	—	—	—	—	—	—	—	—
42	Releasable reduction winch	—	—	—	—	—	—	—	—
43	Baby stroller	—	—	—	—	—	—	—	—
44	Height- and tilt-adjustable worktable	—	—	—	—	—	—	—	—
45	Coin sorter and counter	—	—	—	—	—	—	—	—
46	Interchangeable-tune music box	—	—	—	—	—	—	—	—
Total completed		**1**	**8**	**0**	**22**	**3**	**10**	**8**	**19**

F.3 Verification Scope and Revision Without the Harness

Access to simulation supports meaningful engineering. The simulation-tools trajectories contain active physical diagnosis and geometric repair. For example, Astra adds a detent to an indexing table after identifying motion during the intended dwell phase, then checks recovery from obstructions. Such behavior shows that agents can formulate useful experiments without the harness. The question is how those experiments collectively establish the requested function, and whether that target remains intact as the design and test code change.

Table 15: Testing and revision in the simulation-tools setting. Agent-written checks are distinguished from recorded whole-task completion.

Case	Agent action	Observed evidence	Implication and remaining scope
Task 36 Astra Geneva indexing table	Adds a spring-loaded roller detent after station drift; tests obstructions near entry and exit; relieves slot mouths after a swept-geometry check.	The constructor reports eight 90° steps, dwell holding under disturbances, and recovery after obstruction in its own simulations.	Task-relevant experiments lead to concrete mechanical changes. These local results do not establish whole-task completion in the reported evaluation.
Task 9 Astra Mechanical alarm clock	Tests the escapement for 600 simulated seconds and tests the striker separately; revises gear phasing and local clearances.	The escapement advances approximately ten revolutions. The delivery explicitly leaves the integrated alarm sequence unverified; the hour wheel also interferes with the alarm lever.	Subsystem evidence leaves a composition gap: setting, triggering, and stopping the alarm must all preserve timekeeping. See Fig. 7b.
Task 13 Grok Ratchet screwdriver	Replaces a numerically invalid 1.40-second loaded, two-stroke test with a 0.20-second stepping test.	The new run returns 21 finite samples, but the bit remains nearly stationary. The original drive and return-stroke checks are not repeated.	Restoring executable simulation narrows the tested behavior. The agent identifies the result as a smoke test and does not claim functional success.

Complete functions span interactions and operation sequences. The clock case illustrates why successful component tests do not establish the behavior of an assembled mechanism. The missing evidence concerns interactions between subsystems and the preservation of timekeeping through an alarm sequence. A requirement-level testbed makes these obligations explicit alongside tests of the escapement and striker. This connects functional decomposition to composition: individual tests must jointly cover the user's requested behavior, including transitions between operations.

A revision should preserve the target being tested. The screwdriver case separates numerical recovery from functional recovery. Producing finite simulation states is useful, but it leaves unanswered whether the handle advances the output under load without undoing that motion on return. In a separate full-harness Grok run on Task 13, the second candidate passes both forward-advance and both handle-return checks, while both output-backslip checks remain failures. Although neither run records whole-task completion, the harness result identifies the outstanding requirement after partial progress. These are independently generated candidates; the comparison illustrates how the evaluation target is maintained during revision.

F.4 Diagnosis and Stopping Under Binary Feedback

Binary outcomes shift diagnosis toward additional experiments. A failed test can reflect an incorrect input convention, obstructed motion, or an invalid execution setup. A pass/fail response does not distinguish these explanations. Table 16 shows three responses: designing tests that separate hypotheses, reconstructing evidence through independent geometric reasoning, and revising several plausible failure sources without confirming which caused the rejection.

Table 16: Diagnostic strategies under binary feedback. Geometric and kinematic calculations are the agents' own analyses, not hidden simulation traces.

Case	Agent action	Evidence and outcome	What the case shows
Task 1 Astra Hinged toolbox	Tests rest and opening with opposite torque signs. Reverses the hinge's body ordering after only the reverse-direction opening test passes.	A subsequent positive-torque opening test passes. The complete two-cycle operation remains unverified; the agent delivers partial results.	Carefully chosen binary tests can distinguish an input-direction error from other explanations. Resolving one local cause does not establish the full operating sequence.
Task 12 Grok Nail clipper	Uses its own geometry and linkage calculations to investigate input direction, excessive follower travel, and changes in the four-bar solution branch; revises the mechanism.	The revised design obtains a recorded whole-task pass. Several coupled design changes precede this result.	Restricted feedback still permits successful diagnosis when independent analysis supplies the missing information. This success cannot be attributed to a single parameter change.
Task 33 Grok Spring-latch door lock	Measures clearances and revises hinges, handle bearings, cam orientation, latch geometry, and the strike plate after repeated binary failures.	The agent finds geometric problems, but the returned failures do not confirm which defect caused rejection. No whole-task completion is recorded.	Useful inspection can coexist with unresolved failure localization. The full-harness repair in Fig. 8 instead tracks relocking checks and then residual interference.

The models also differ in when they stop. Table 17 summarizes the recorded termination states in the binary-feedback batch. Astra chooses partial delivery in 43 of 46 runs, with a median of 137.8 minutes of active budget remaining at those decisions. Its explanations repeatedly cite insufficient evidence for a targeted next revision. Grok instead reaches the budget limit in 36 runs and completes eight. The same restricted feedback thus accompanies different search and stopping strategies; Astra's zero recorded completions should not be read as 46 fully executed mechanical failures.

Table 17: Recorded termination states under binary feedback, with 46 runs per model. Categories are mutually exclusive. Evidence required denotes a recorded unresolved evidence state.

Model	Completed	Partial delivery	Budget exhausted	Evidence required
Astra	0	43	0	3
Grok	8	0	36	2

Together, these observations separate a correctness signal from evidence for choosing a revision. Binary outcomes can guide repair through additional experiments or independent analysis. Detailed feedback supplies further information about the failed operation or validity condition, helping the agent select and verify its next change. The stopping counts compare models within the binary setting; estimating how masking changes each model's stopping policy would require a paired analysis of the full-harness runs.

F.5 Construction Without Visual Access

Section 6 notes that agents rarely inspect rendered images during functional design. To test whether visual access nonetheless contributes to task success, we repeat the single-attempt Astra full-harness experiment with one additional restriction on the constructor. Task prompts, constructor instructions, the harness workflow, physical simulation, diagnostic numerical feedback, the active-time budget, and the independent final evaluation are identical to Section 5.1. The constructor, however, cannot view, generate, or capture images, and cannot operate a computer or browser through screenshots. It cannot render the model through the harness preview tool, through its own scripts using plotting, imaging, or CAD visualization libraries, or through external modeling and rendering programs. It also cannot open image, video, 3D-viewer, HTML, or PDF files, and harness feedback returned to it contains only text and numerical values. The constructor retains its CAD source and exported geometry (STEP and STL), programmatic geometric queries, and simulation. The test-binding agent, independent review, and final evaluation keep their standard visual access, so the restriction removes visual information only from design and revision.

Table 18 compares the two runs. Without visual access, Astra completes 20 of the 46 tasks, compared with 22 with visual access. Sixteen tasks are solved in both settings. Six are solved only with visual access: the four-bar wiper, piston pump, two-digit counter, gravity-pawl turntable, tilting mirror, and folding shopping cart (Tasks 3, 17, 19, 23, 28, and 37). Four are solved only without it: the hinged toolbox, scissor lift, toggle clamp, and parallel gripper (Tasks 1, 14, 16, and 41).

Removing visual access therefore changes the aggregate count by two tasks, while the sets of solved tasks differ in ten. With one attempt per task, these task-level differences cannot be attributed to visual access alone. The aggregate result nevertheless indicates that, within FunCAD-Harness, the constructor's functional success depends mainly on CAD programs, geometric queries, and simulation feedback rather than on visual inspection, consistent with the tool-use profiles in Appendix E.6.

Table 18: Task-level recorded completion of single-attempt Astra full-harness runs with and without visual access for the constructor. A check mark denotes completion under formal evaluation; a dash denotes no recorded completion, including unfinished evaluation. Each setting has one independent attempt per task; the *With vision* column reproduces the Astra column of Table 8.

ID	Mechanism	With vision	No vision
1	Hinged toolbox	—	✓
2	Rack-and-pinion gate	✓	✓
3	Four-bar wiper	✓	—
4	Hand-cranked egg beater	✓	✓
5	Self-locking utility knife	✓	✓
6	Folding drying rack	—	—
7	Bicycle	—	—
8	Pedal-operated bin	✓	✓
9	Mechanical alarm clock	—	—
10	Kitchen tongs	—	—
11	Retractable ballpoint pen	—	—
12	Nail clipper	✓	✓
13	One-way ratchet screwdriver	—	—
14	Linearly actuated scissor lift	—	✓
15	Equal-arm balance	✓	✓
16	Toggle clamp / press	—	✓
17	Dry piston-pump mechanism	✓	—
18	Three-digit combination lock	✓	✓
19	Two-digit mechanical counter	✓	—
20	Three-ball dispenser	—	—
21	Gravity timer	✓	✓
22	Slider-crank push rod	✓	✓
23	Gravity-pawl turntable	✓	—
24	Sliding-bolt gate	✓	✓
25	Three-stage gear reducer	—	—

Continued on next page

Table 18 continued

ID	Mechanism	With vision	No vision
26	Desktop display turntable	✓	✓
27	Small swing gate	✓	✓
28	Tilting desktop mirror	✓	—
29	Folding wall hook	✓	✓
30	Spring clothespin	—	—
31	Scissors	✓	✓
32	Ratchet hand winch	—	—
33	Spring-latch door lock	✓	✓
34	Reversible dual-pawl ratchet wrench	—	—
35	Full-cycle ratchet crimper	—	—
36	Geneva indexing table with overload clutch	—	—
37	Folding shopping cart	✓	—
38	Mechanical 4×4 puzzle cube	—	—
39	Twin-cylinder engine linkage	✓	✓
40	Four-station indexing table	—	—
41	Synchronized parallel gripper	—	✓
42	Releasable reduction winch	—	—
43	Baby stroller	—	—
44	Height- and tilt-adjustable worktable	—	—
45	Coin sorter and counter	—	—
46	Interchangeable-tune music box	—	—
Total completed		**22/46**	**20/46**

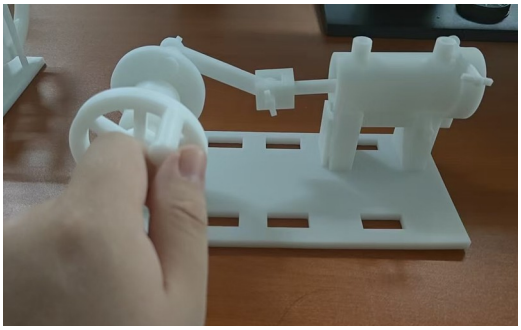
G 3D-Printed Mechanical Prototypes

To illustrate the physical realization of AI-generated mechanical designs, we 3D-printed and assembled six prototypes. Figures 12 and 13 show the resulting piston mechanism, two-digit counter, gravity timer, combination lock, ball distributor, and cup-access machine. These examples provide qualitative evidence of fabrication and assembly, complementing the simulation-based evaluation. Task numbers follow Appendix B; the cup-access machine is the open-ended example in Figure 1. Supplementary videos show operation of the printed mechanisms.

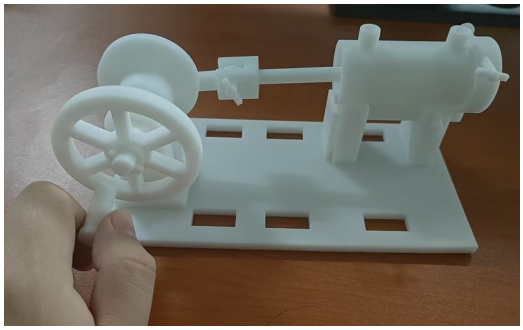
Piston (Task 17). The crank and connecting rod drive a guided piston back and forth. The task requires repeated strokes with the specified travel and return position, including motion against resistance. Panels (a–b) show the printed assembly at two crank positions.

Combination lock (Task 18). Three independently adjustable wheels encode the combination. The task requires unlocking at the correct code, re-locking after reinserting the locking bar and scrambling the wheels, and blocking withdrawal when any one digit is incorrect.

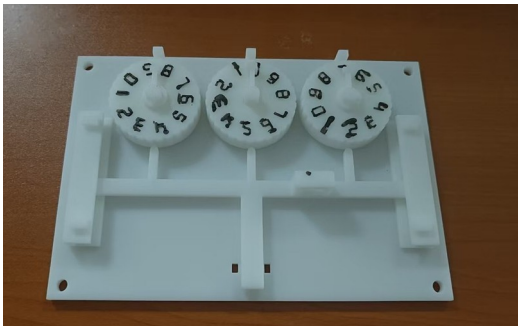
Ball distributor (Task 20). The slider meters balls from the loading channels into the receiving tray. The task requires three balls per cycle, cumulative outputs of 3, 6, and 9 across three cycles, and retention of the remaining balls between operations.



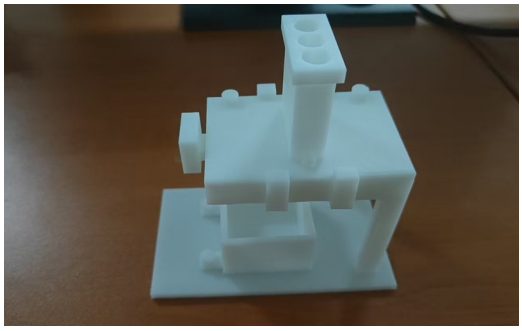
(a) Piston: crank position 1



(b) Piston: crank position 2

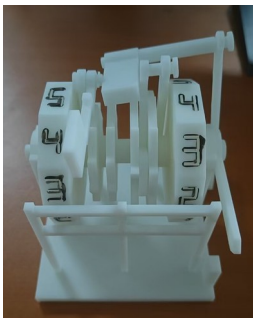


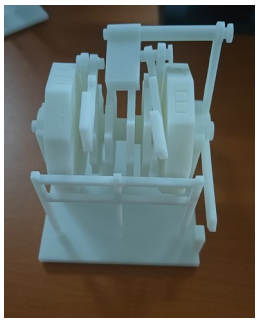
(c) Three-digit combination lock



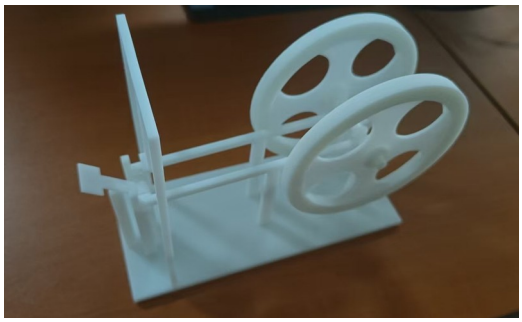
(d) Three-ball dispenser

Figure 12: **Printed prototypes for piston motion, code-controlled locking, and ball dispensing.** (a–b) Task 17: two crank positions of the piston linkage. (c) Task 18: three numbered code wheels and the locking bar. (d) Task 20: the loading channels, dispensing slider, and receiving tray.





(a) Two-digit counter: two views



(b) Gravity timer



(c) Cup-access machine: initial configuration

Figure 13: **Printed prototypes for counting, timing, and cup access.** (a) Task 19: two views of the counter, showing its digit wheels and actuating linkage. (b) Task 21: the gravity-timer assembly. (c) The open-ended cup-access mechanism with its target cup at the rear of the arrangement. The pen identifies the target cup; the raised configuration and cup retrieval appear in Figure 1.

Two-digit counter (Task 19). The task requires one increment per forward push-rod stroke, correct carry through $09 \rightarrow 10$ and $19 \rightarrow 20$, and retention of the reading during the return stroke and at rest. The two views expose the digit wheels and the linkage that advances them.

Gravity timer (Task 21). The prototype implements a nominal 10-second gravity timer. The task requires autonomous operation after release, triggering within the specified time window, retention of the visible indication, and repeat operation after reset.

Cup-access machine (open-ended task). The design addresses access to a cup stored behind other cups. Turning the handle moves the paired side linkages and raises the cup holder into reach. Panel (c) shows the initial arrangement; the supplementary video shows operation of the printed linkage and cup retrieval.